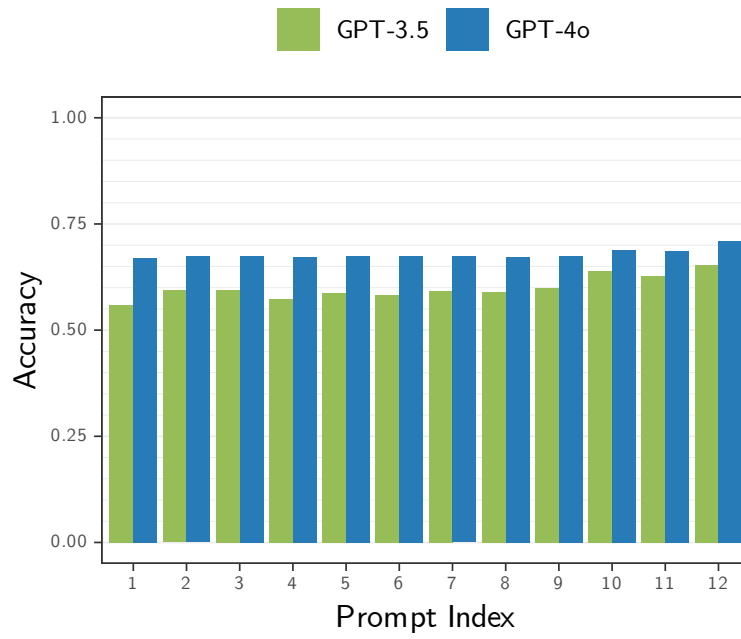


# Results

Aug 12, 2024

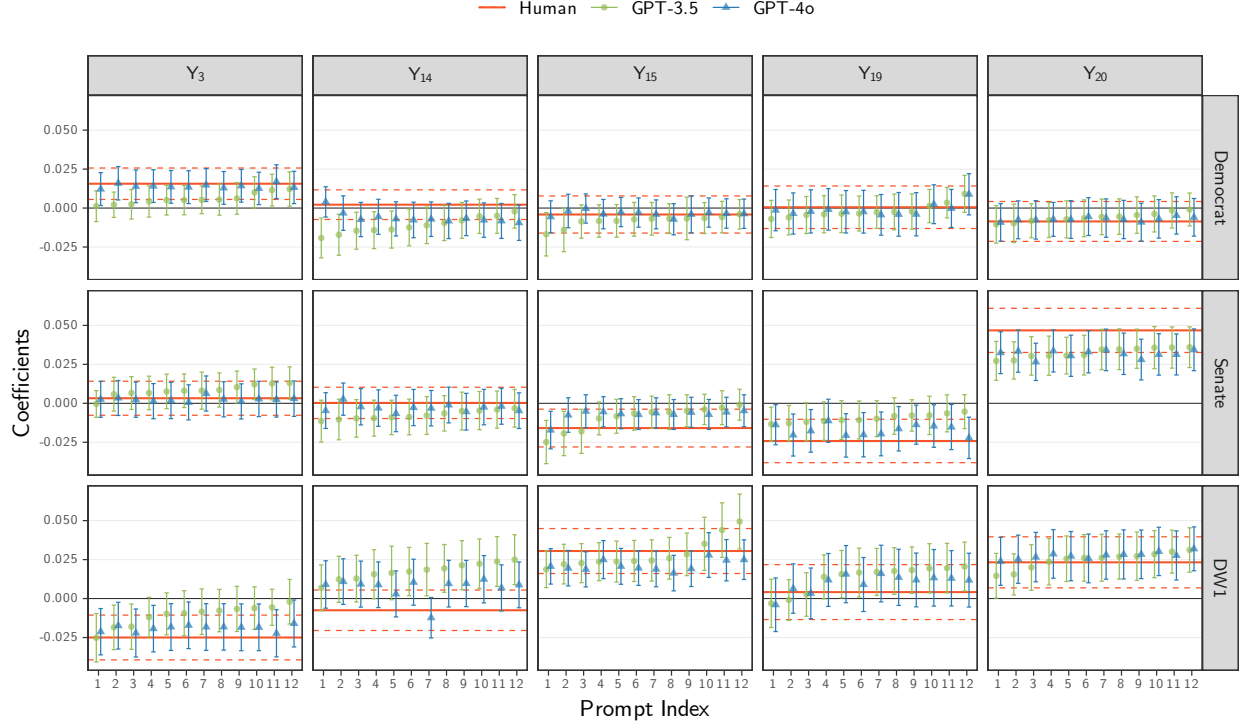
Figure 1: Accuracy of Topic Predictions vs. Prompt.



Note:  $\text{Accuracy} = \frac{1}{10,000} \sum_{i=1}^{10,000} 1\{Y_{m,p}^{\text{LLM}} = Y^{\text{Human}}\}^{(i)}$

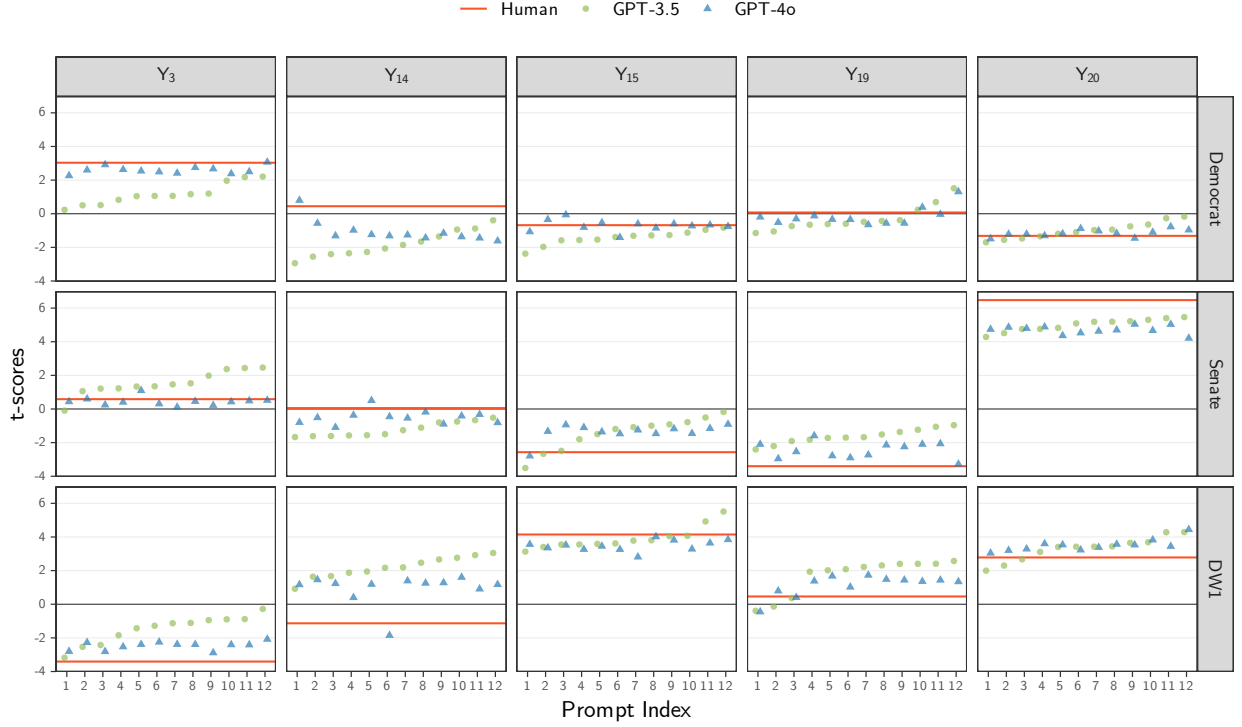
## Proxy on the LHS

Figure 2:  $\beta_1$  vs. Prompt. Proxy on the LHS.



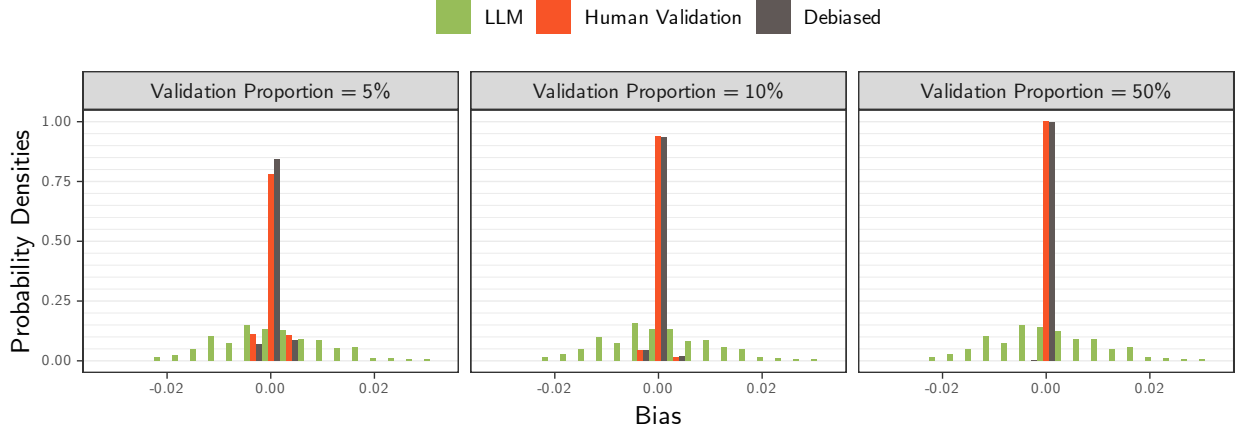
Note: The coefficients were obtained by fitting the model  $Y_t = \beta_0 + \beta_1 V$  over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here,  $Y_t$  denotes the indicator for topic  $t$ , where  $Y_t = 1\{Y = t\}$  for  $t \in \mathcal{T}$  and  $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$ , with  $\mathcal{T} = \{3, 14, 15, 19, 20\}$  is the 5 most-common human-labeled topics in the bills data. The variable  $V$  takes one of the following bills variables: Democrat, Senate, or DW1, where  $\text{extDemocrat} = 1\{\text{Party} = \text{Democrat}\}$ ,  $\text{Senate} = 1\{\text{Chamber} = \text{Senate}\}$ , and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with  $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$ , while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with  $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$ , where  $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$  denotes the LLM model and  $p \in \{1, \dots, 12\}$  denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 3: t-scores of  $\beta_1$  vs. Prompt. Proxy on the LHS.



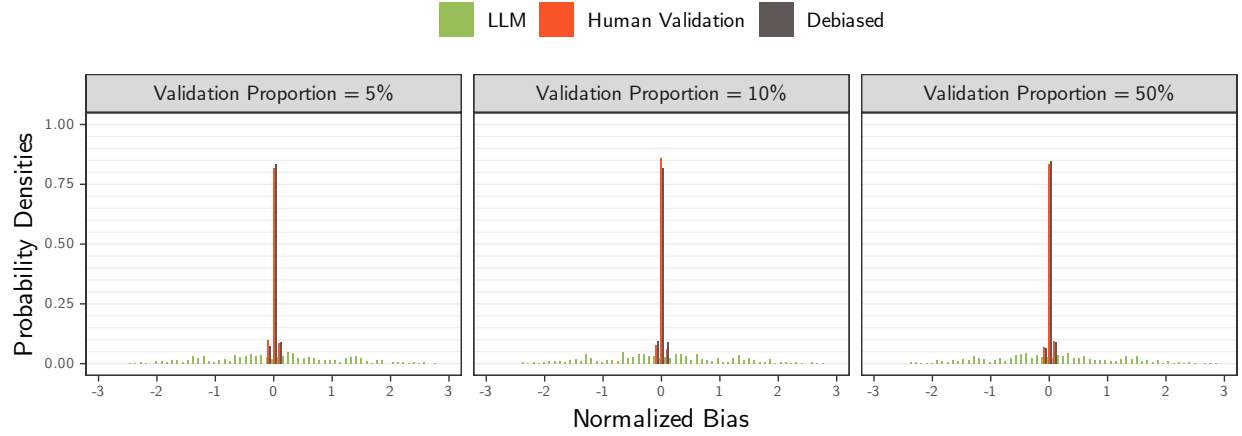
Note: The t-scores were obtained by fitting the model  $Y_t = \beta_0 + \beta_1 V$  over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here,  $Y_t$  denotes the indicator for topic  $t$ , where  $Y_t = 1\{Y = t\}$  for  $t \in \mathcal{T}$  and  $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$ , with  $\mathcal{T} = \{3, 14, 15, 19, 20\}$  is the 5 most-common human-labeled topics in the bills data. The variable  $V$  takes one of the following bills variables: Democrat, Senate, or DW1, where  $\text{extDemocrat} = 1\{\text{Party} = \text{Democrat}\}$ ,  $\text{Senate} = 1\{\text{Chamber} = \text{Senate}\}$ , and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with  $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$ , while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with  $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$ , where  $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$  denotes the LLM model and  $p \in \{1, \dots, 12\}$  denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

Figure 4: Bias Density of  $\beta_1$  by Proportion of Validation Samples. Proxy on the LHS.



Note: The bias estimates were computed as the difference between the coefficient estimates from  $N = 1000$  simulation runs,  $\bar{\beta}_{t,V,\eta,m,p} = \sum_{i=1}^N \beta_{t,V,\eta,m,p}^{(i)} / N$ , and the human-based coefficient from the entire set of 10,000 bills,  $\beta_{t,V}^*$ , using a regression model of the form  $Y_t = \beta_0 + \beta_1 V$ , for topic  $t \in \{3, 14, 15, 19, 20, \text{Other}\}$ , variable  $V \in \{\text{Democrat, Senate, DW1}\}$ , validation proportion  $\eta \in \{5\%, 10\%, 25\%, 50\%\}$ , LLM model  $m \in \{\text{GPT-3.5, GPT-4o}\}$ , and prompt  $p \in \{1, \dots, 12\}$ . For the LLM-based models (green), a random sample of 5,000 bills drawn from the original 10,000 was used, with LLM-based labels,  $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$ . For the other two models—human-based (red) and debiased-based (brown)—the 5,000 sample were split into two subsets: validation and primary, with the size of the validation set determined by the validation proportion,  $\eta$ . The results for  $\eta = 25\%$  were excluded from the figures for clarity. In the human-based model, only the human labels in the validation set were used,  $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$ , whereas the debiased estimates use both the validation and primary sets, with  $Y_t^{\text{Human}}$  from validation set used to estimate measurement errors and  $Y_{t,m,p}^{\text{LLM}}$  from the primary set used to obtain the coefficient estimates used to compute the reported bias.

Figure 5: Normalized Bias Density of  $\beta_1$  by Proportion of Validation Samples. Proxy on the LHS.



Note: The normalized bias estimates were computed as the normalized difference between the coefficient estimates from  $N = 1000$  simulation runs,  $\bar{\beta}_{t,V,\eta,m,p}$ , and the human-based coefficient from the entire set of 10,000 bills,  $\beta_{t,V}^*$ , divided by the sample standard deviation,  $s_{t,V,\eta,m,p}$ , where  $s_{t,V,\eta,m,p}^2 = \sum_{i=1}^N (\beta_{t,V,\eta,m,p}^{(i)} - \bar{\beta}_{t,V,\eta,m,p})^2 / (N - 1)$  and all variables are as defined previously.

Figure 6: Bias Density of  $\beta_1$  by Model and Proportion of Validation Samples. Proxy on the LHS.

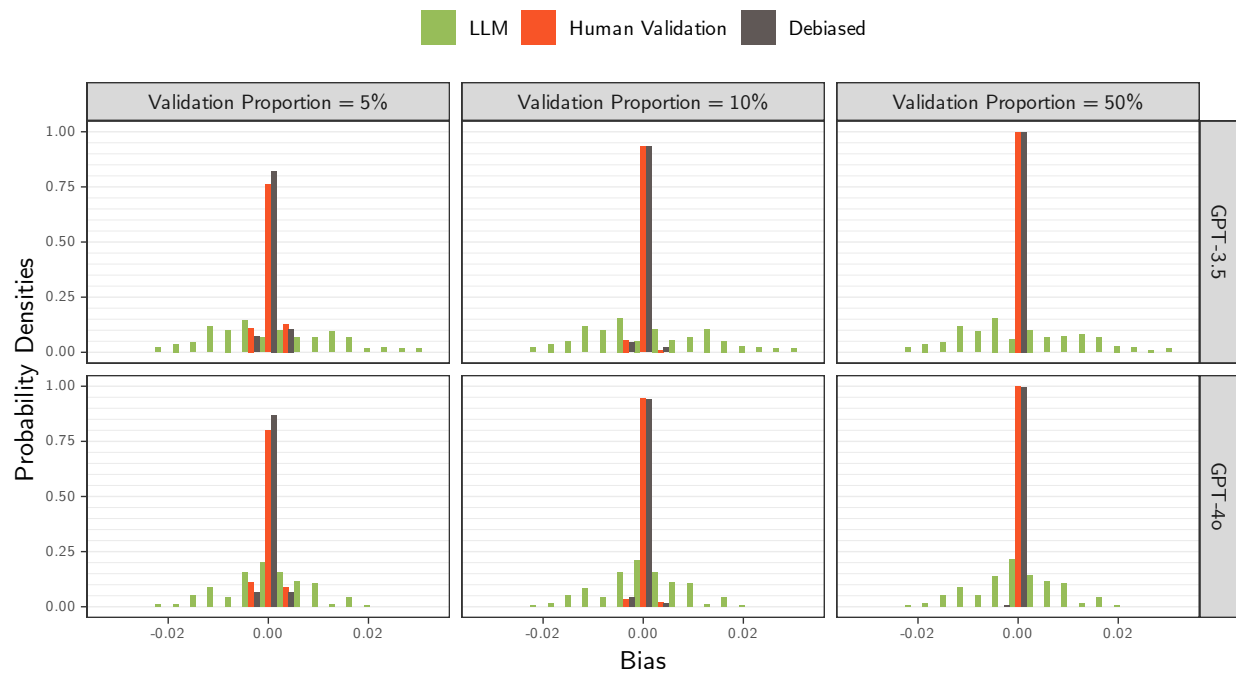


Figure 7: Normalized Bias Density of  $\beta_1$  by Model and Proportion of Validation Samples. Proxy on the LHS.



Figure 8: MSE Empirical CDF of  $\beta_1$  by Proportion of Validation Samples. Proxy on the LHS.

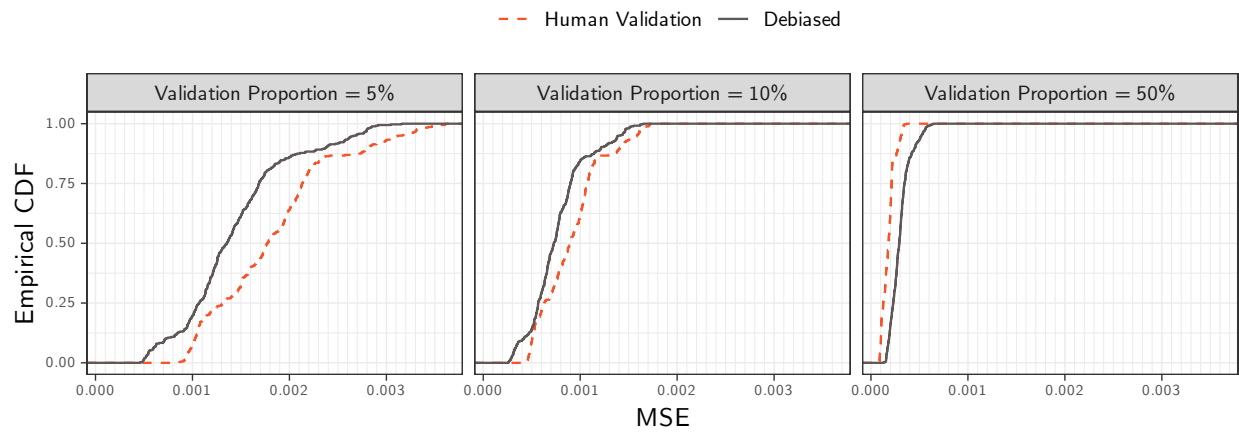




Figure 9: MSE Empirical CDF of  $\beta_1$  by Model and Proportion of Validation Samples. Proxy on the LHS.

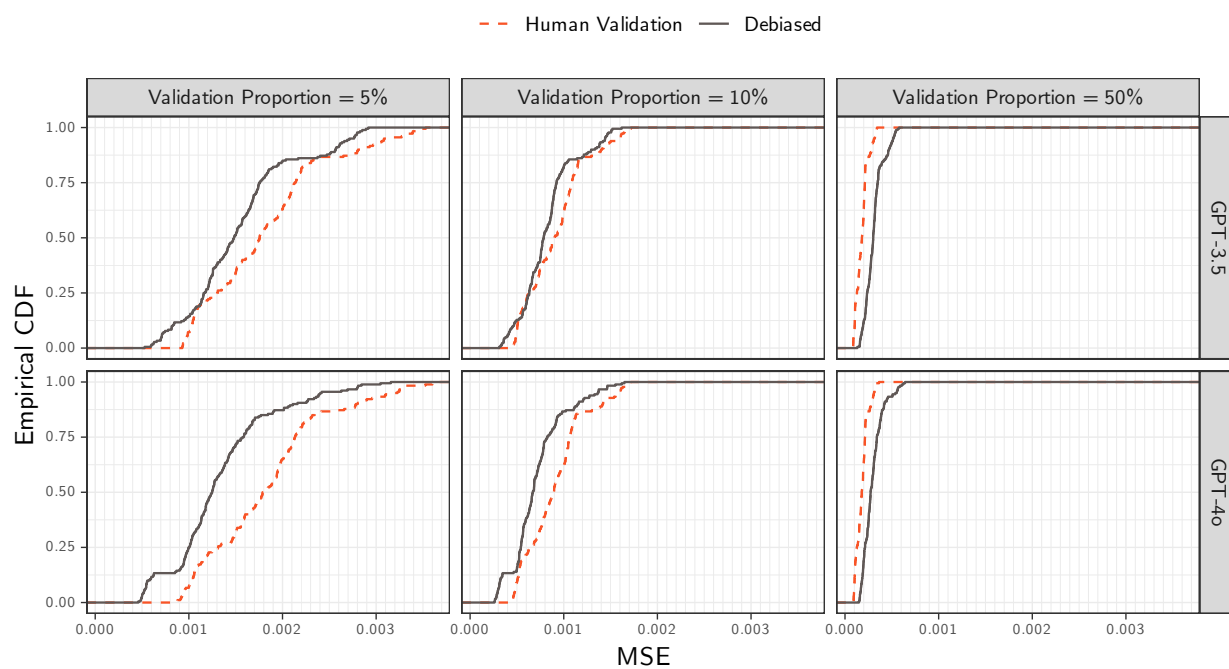


Table 1: Bias Summary Statistics of  $\beta_1$  by Proportion of Validation Samples. Proxy on the LHS:  $Y = \beta_0 + \beta_1 V$ .

| Validation Proportion | Bias             | Mean  | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|-------|-------|--------|--------|-------|
| 5%                    | LLM              | 0.001 | 0.010 | 0.001  | -0.015 | 0.018 |
|                       | Human Validation | 0.000 | 0.001 | 0.000  | -0.002 | 0.002 |
|                       | Debiased         | 0.000 | 0.001 | 0.000  | -0.002 | 0.002 |
| 10%                   | LLM              | 0.001 | 0.010 | 0.000  | -0.015 | 0.018 |
|                       | Human Validation | 0.000 | 0.001 | 0.000  | -0.002 | 0.001 |
|                       | Debiased         | 0.000 | 0.001 | 0.000  | -0.002 | 0.001 |
| 25%                   | LLM              | 0.001 | 0.010 | 0.000  | -0.014 | 0.018 |
|                       | Human Validation | 0.000 | 0.001 | 0.000  | -0.001 | 0.001 |
|                       | Debiased         | 0.000 | 0.001 | 0.000  | -0.001 | 0.001 |
| 50%                   | LLM              | 0.001 | 0.010 | 0.000  | -0.014 | 0.019 |
|                       | Human Validation | 0.000 | 0.000 | 0.000  | -0.001 | 0.001 |
|                       | Debiased         | 0.000 | 0.001 | 0.000  | -0.001 | 0.001 |

Table 2: Normalized Bias Summary Statistics of  $\beta_1$  by Proportion of Validation Samples. Proxy on the LHS:  $Y = \beta_0 + \beta_1 V$ .

| Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|--------|-------|--------|--------|-------|
| 5%                    | LLM              | 0.071  | 1.099 | 0.067  | -1.692 | 1.856 |
|                       | Human Validation | 0.000  | 0.034 | 0.001  | -0.055 | 0.054 |
|                       | Debiased         | 0.003  | 0.033 | 0.001  | -0.055 | 0.061 |
| 10%                   | LLM              | 0.069  | 1.094 | 0.040  | -1.722 | 1.869 |
|                       | Human Validation | -0.002 | 0.030 | -0.001 | -0.055 | 0.048 |
|                       | Debiased         | -0.002 | 0.033 | -0.002 | -0.060 | 0.049 |
| 25%                   | LLM              | 0.073  | 1.099 | 0.023  | -1.711 | 1.854 |
|                       | Human Validation | 0.002  | 0.031 | 0.001  | -0.050 | 0.051 |
|                       | Debiased         | 0.001  | 0.033 | 0.001  | -0.049 | 0.057 |
| 50%                   | LLM              | 0.076  | 1.096 | 0.047  | -1.654 | 1.875 |
|                       | Human Validation | 0.004  | 0.032 | 0.003  | -0.051 | 0.056 |
|                       | Debiased         | 0.000  | 0.031 | -0.001 | -0.050 | 0.053 |

Table 3: Bias Summary Statistics by Model and Proportion of Validation Samples for  $\beta_1$ . Proxy on the LHS:  $Y = \beta_0 + \beta_1 V$ .

| Model   | Validation Proportion | Bias             | Mean  | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.002 | 0.012 | 0.000  | -0.016 | 0.023 |
| GPT-3.5 | 5%                    | Human Validation | 0.000 | 0.001 | 0.000  | -0.002 | 0.003 |
| GPT-3.5 | 5%                    | Debiased         | 0.000 | 0.001 | 0.000  | -0.002 | 0.002 |
| GPT-3.5 | 10%                   | LLM              | 0.002 | 0.012 | 0.000  | -0.016 | 0.023 |
| GPT-3.5 | 10%                   | Human Validation | 0.000 | 0.001 | 0.000  | -0.002 | 0.001 |
| GPT-3.5 | 10%                   | Debiased         | 0.000 | 0.001 | 0.000  | -0.002 | 0.001 |
| GPT-3.5 | 25%                   | LLM              | 0.002 | 0.012 | 0.000  | -0.016 | 0.023 |
| GPT-3.5 | 25%                   | Human Validation | 0.000 | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-3.5 | 25%                   | Debiased         | 0.000 | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-3.5 | 50%                   | LLM              | 0.002 | 0.012 | 0.000  | -0.016 | 0.022 |
| GPT-3.5 | 50%                   | Human Validation | 0.000 | 0.000 | 0.000  | -0.001 | 0.001 |
| GPT-3.5 | 50%                   | Debiased         | 0.000 | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-4o  | 5%                    | LLM              | 0.001 | 0.008 | 0.001  | -0.013 | 0.015 |
| GPT-4o  | 5%                    | Human Validation | 0.000 | 0.001 | 0.000  | -0.002 | 0.002 |
| GPT-4o  | 5%                    | Debiased         | 0.000 | 0.001 | 0.000  | -0.002 | 0.002 |
| GPT-4o  | 10%                   | LLM              | 0.001 | 0.008 | 0.001  | -0.014 | 0.014 |
| GPT-4o  | 10%                   | Human Validation | 0.000 | 0.001 | 0.000  | -0.002 | 0.001 |
| GPT-4o  | 10%                   | Debiased         | 0.000 | 0.001 | 0.000  | -0.002 | 0.001 |
| GPT-4o  | 25%                   | LLM              | 0.001 | 0.008 | 0.000  | -0.013 | 0.014 |
| GPT-4o  | 25%                   | Human Validation | 0.000 | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-4o  | 25%                   | Debiased         | 0.000 | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-4o  | 50%                   | LLM              | 0.001 | 0.008 | 0.001  | -0.013 | 0.015 |
| GPT-4o  | 50%                   | Human Validation | 0.000 | 0.000 | 0.000  | -0.001 | 0.001 |
| GPT-4o  | 50%                   | Debiased         | 0.000 | 0.001 | 0.000  | -0.001 | 0.001 |

Table 4: Normalized Bias Summary Statistics by Model and Proportion of Validation Samples for  $\beta_1$ . Proxy on the LHS:  $Y = \beta_0 + \beta_1 V$ .

| Model   | Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|--------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.113  | 1.284 | -0.023 | -1.899 | 2.211 |
| GPT-3.5 | 5%                    | Human Validation | 0.001  | 0.034 | 0.003  | -0.052 | 0.056 |
| GPT-3.5 | 5%                    | Debiased         | 0.003  | 0.034 | 0.001  | -0.055 | 0.066 |
| GPT-3.5 | 10%                   | LLM              | 0.109  | 1.280 | -0.005 | -1.863 | 2.179 |
| GPT-3.5 | 10%                   | Human Validation | -0.003 | 0.031 | -0.001 | -0.064 | 0.042 |
| GPT-3.5 | 10%                   | Debiased         | -0.003 | 0.032 | -0.005 | -0.053 | 0.049 |
| GPT-3.5 | 25%                   | LLM              | 0.118  | 1.285 | 0.007  | -1.837 | 2.264 |
| GPT-3.5 | 25%                   | Human Validation | 0.004  | 0.029 | 0.002  | -0.046 | 0.050 |
| GPT-3.5 | 25%                   | Debiased         | 0.002  | 0.031 | 0.002  | -0.044 | 0.053 |
| GPT-3.5 | 50%                   | LLM              | 0.116  | 1.282 | -0.004 | -1.864 | 2.172 |
| GPT-3.5 | 50%                   | Human Validation | 0.004  | 0.033 | 0.003  | -0.052 | 0.060 |
| GPT-3.5 | 50%                   | Debiased         | 0.000  | 0.030 | 0.000  | -0.050 | 0.049 |
| GPT-4o  | 5%                    | LLM              | 0.029  | 0.877 | 0.084  | -1.411 | 1.514 |
| GPT-4o  | 5%                    | Human Validation | -0.001 | 0.033 | 0.000  | -0.055 | 0.051 |
| GPT-4o  | 5%                    | Debiased         | 0.002  | 0.031 | 0.001  | -0.055 | 0.054 |
| GPT-4o  | 10%                   | LLM              | 0.028  | 0.871 | 0.059  | -1.447 | 1.507 |
| GPT-4o  | 10%                   | Human Validation | -0.001 | 0.030 | -0.002 | -0.053 | 0.049 |
| GPT-4o  | 10%                   | Debiased         | -0.001 | 0.034 | 0.000  | -0.066 | 0.050 |
| GPT-4o  | 25%                   | LLM              | 0.027  | 0.876 | 0.056  | -1.422 | 1.463 |
| GPT-4o  | 25%                   | Human Validation | 0.001  | 0.032 | -0.001 | -0.050 | 0.052 |
| GPT-4o  | 25%                   | Debiased         | 0.000  | 0.035 | -0.001 | -0.053 | 0.060 |
| GPT-4o  | 50%                   | LLM              | 0.035  | 0.874 | 0.070  | -1.441 | 1.510 |
| GPT-4o  | 50%                   | Human Validation | 0.004  | 0.031 | 0.003  | -0.050 | 0.051 |
| GPT-4o  | 50%                   | Debiased         | 0.000  | 0.031 | -0.001 | -0.045 | 0.059 |

Table 5: MSE Summary Statistics of  $\beta_1$  by Proportion of Validation Samples. Proxy on the LHS:  $Y = \beta_0 + \beta_1 V$ .

| Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|-----------------------|------------------|-------|-------|--------|-------|-------|
| 5%                    | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| 5%                    | Human Validation | 0.002 | 0.001 | 0.002  | 0.001 | 0.003 |
| 5%                    | Debiased         | 0.001 | 0.001 | 0.001  | 0.001 | 0.003 |
| 10%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| 10%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.000 | 0.002 |
| 10%                   | Debiased         | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| 25%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| 25%                   | Human Validation | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| 25%                   | Debiased         | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| 50%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| 50%                   | Human Validation | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| 50%                   | Debiased         | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |

Table 6: MSE Summary Statistics by Model and Proportion of Validation Samples for  $\beta_1$ . Proxy on the LHS:  $Y = \beta_0 + \beta_1 V$ .

| Model   | Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|-------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| GPT-3.5 | 5%                    | Human Validation | 0.002 | 0.001 | 0.002  | 0.001 | 0.003 |
| GPT-3.5 | 5%                    | Debiased         | 0.002 | 0.001 | 0.001  | 0.001 | 0.003 |
| GPT-3.5 | 10%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| GPT-3.5 | 10%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.000 | 0.002 |
| GPT-3.5 | 10%                   | Debiased         | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 25%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| GPT-3.5 | 25%                   | Human Validation | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| GPT-3.5 | 25%                   | Debiased         | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| GPT-3.5 | 50%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| GPT-3.5 | 50%                   | Human Validation | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-3.5 | 50%                   | Debiased         | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| GPT-4o  | 5%                    | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-4o  | 5%                    | Human Validation | 0.002 | 0.001 | 0.002  | 0.001 | 0.003 |
| GPT-4o  | 5%                    | Debiased         | 0.001 | 0.001 | 0.001  | 0.001 | 0.002 |
| GPT-4o  | 10%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-4o  | 10%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.000 | 0.002 |
| GPT-4o  | 10%                   | Debiased         | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 25%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-4o  | 25%                   | Human Validation | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| GPT-4o  | 25%                   | Debiased         | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| GPT-4o  | 50%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-4o  | 50%                   | Human Validation | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-4o  | 50%                   | Debiased         | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |

Table 7: Coverage Summary Statistics of  $\beta_1$  by Proportion of Validation Samples. Proxy on the LHS:  $Y = \beta_0 + \beta_1 V$ .

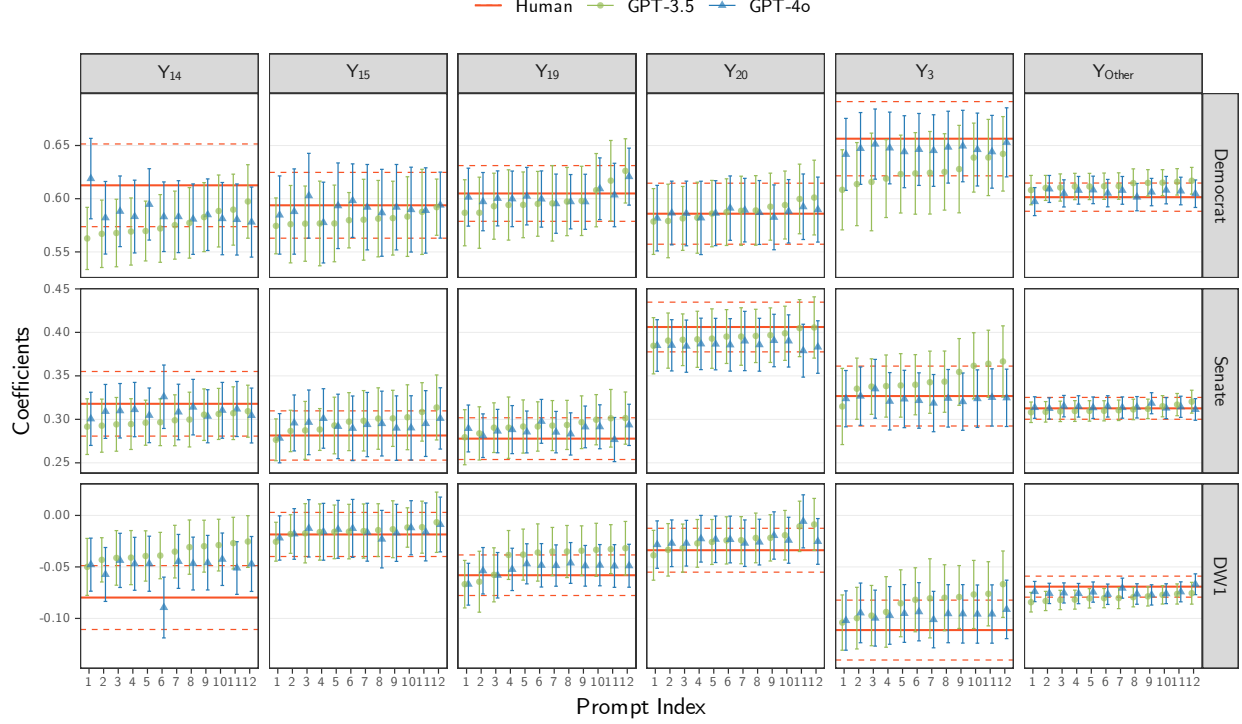
| Validation Proportion |     | Coverage         | Mean  | SD    | Median | 5%    | 95%   |
|-----------------------|-----|------------------|-------|-------|--------|-------|-------|
|                       | 5%  | LLM              | 0.811 | 0.163 | 0.889  | 0.461 | 0.951 |
|                       | 5%  | Human Validation | 0.946 | 0.008 | 0.946  | 0.932 | 0.959 |
|                       | 5%  | Debiased         | 0.928 | 0.012 | 0.929  | 0.904 | 0.945 |
|                       | 10% | LLM              | 0.810 | 0.165 | 0.889  | 0.460 | 0.951 |
|                       | 10% | Human Validation | 0.948 | 0.007 | 0.949  | 0.936 | 0.959 |
|                       | 10% | Debiased         | 0.940 | 0.008 | 0.941  | 0.926 | 0.952 |
|                       | 25% | LLM              | 0.809 | 0.164 | 0.887  | 0.465 | 0.952 |
|                       | 25% | Human Validation | 0.949 | 0.007 | 0.949  | 0.936 | 0.960 |
|                       | 25% | Debiased         | 0.946 | 0.007 | 0.946  | 0.934 | 0.957 |
|                       | 50% | LLM              | 0.810 | 0.164 | 0.887  | 0.463 | 0.950 |
|                       | 50% | Human Validation | 0.950 | 0.007 | 0.950  | 0.939 | 0.961 |
|                       | 50% | Debiased         | 0.948 | 0.008 | 0.948  | 0.935 | 0.959 |

Table 8: Coverage Summary Statistics by Model and Proportion of Validation Samples for  $\beta_1$ . Proxy on the LHS:  $Y = \beta_0 + \beta_1 V$ .

| Model   | Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|-------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.761 | 0.187 | 0.819  | 0.381 | 0.945 |
| GPT-3.5 | 5%                    | Human Validation | 0.946 | 0.008 | 0.948  | 0.933 | 0.961 |
| GPT-3.5 | 5%                    | Debiased         | 0.929 | 0.011 | 0.931  | 0.910 | 0.945 |
| GPT-3.5 | 10%                   | LLM              | 0.759 | 0.189 | 0.812  | 0.383 | 0.949 |
| GPT-3.5 | 10%                   | Human Validation | 0.947 | 0.007 | 0.947  | 0.936 | 0.959 |
| GPT-3.5 | 10%                   | Debiased         | 0.940 | 0.008 | 0.941  | 0.927 | 0.952 |
| GPT-3.5 | 25%                   | LLM              | 0.760 | 0.188 | 0.815  | 0.362 | 0.945 |
| GPT-3.5 | 25%                   | Human Validation | 0.949 | 0.007 | 0.949  | 0.936 | 0.958 |
| GPT-3.5 | 25%                   | Debiased         | 0.946 | 0.007 | 0.946  | 0.934 | 0.957 |
| GPT-3.5 | 50%                   | LLM              | 0.760 | 0.190 | 0.815  | 0.369 | 0.950 |
| GPT-3.5 | 50%                   | Human Validation | 0.951 | 0.007 | 0.950  | 0.939 | 0.962 |
| GPT-3.5 | 50%                   | Debiased         | 0.947 | 0.008 | 0.948  | 0.934 | 0.959 |
| GPT-4o  | 5%                    | LLM              | 0.861 | 0.116 | 0.921  | 0.637 | 0.954 |
| GPT-4o  | 5%                    | Human Validation | 0.945 | 0.008 | 0.946  | 0.930 | 0.957 |
| GPT-4o  | 5%                    | Debiased         | 0.926 | 0.014 | 0.927  | 0.902 | 0.945 |
| GPT-4o  | 10%                   | LLM              | 0.860 | 0.117 | 0.919  | 0.642 | 0.952 |
| GPT-4o  | 10%                   | Human Validation | 0.949 | 0.007 | 0.949  | 0.936 | 0.959 |
| GPT-4o  | 10%                   | Debiased         | 0.940 | 0.008 | 0.941  | 0.926 | 0.953 |
| GPT-4o  | 25%                   | LLM              | 0.859 | 0.117 | 0.921  | 0.625 | 0.954 |
| GPT-4o  | 25%                   | Human Validation | 0.949 | 0.007 | 0.949  | 0.936 | 0.960 |
| GPT-4o  | 25%                   | Debiased         | 0.946 | 0.007 | 0.946  | 0.934 | 0.958 |
| GPT-4o  | 50%                   | LLM              | 0.860 | 0.115 | 0.919  | 0.635 | 0.950 |
| GPT-4o  | 50%                   | Human Validation | 0.950 | 0.007 | 0.949  | 0.939 | 0.961 |
| GPT-4o  | 50%                   | Debiased         | 0.948 | 0.007 | 0.948  | 0.935 | 0.959 |

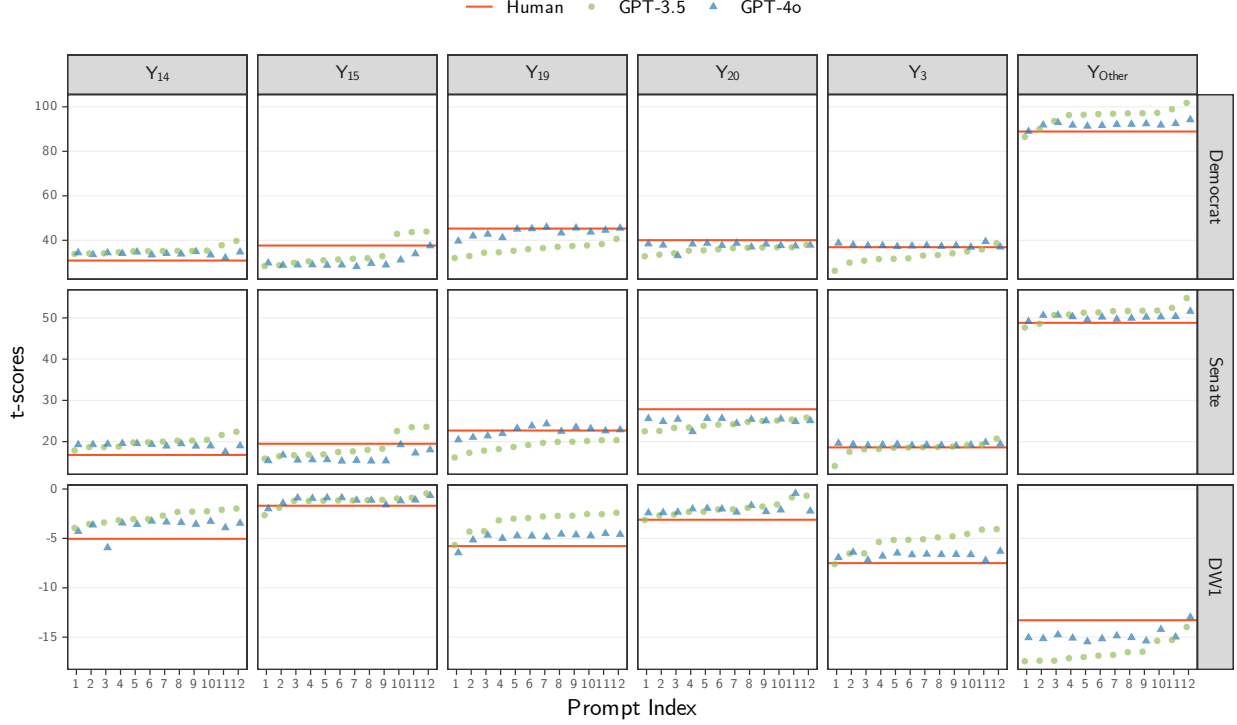
## Proxy on the RHS

Figure 10:  $\beta_t$  vs. Prompt for all  $\beta$ s. Proxy on the RHS.



Note: The coefficients were obtained by fitting the model  $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{Other}\beta_{Other}$  over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here,  $Y_t$  denotes the indicator for topic  $t$ , where  $Y_t = 1\{Y = t\}$  for  $t \in \mathcal{T}$  and  $Y_{Other} = 1\{Y \notin \mathcal{T}\}$ , with  $\mathcal{T} = \{3, 14, 15, 19, 20\}$  is the 5 most-common human-labeled topics in the bills data. The variable  $V$  takes one of the following bills variables: Democrat, Senate, or DW1, where  $extDemocrat = 1\{\text{Party} = \text{Democrat}\}$ ,  $Senate = 1\{\text{Chamber} = \text{Senate}\}$ , and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with  $Y_t^{Human} = 1\{Y^{Human} = t\}$ , while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with  $Y_{t,m,p}^{LLM} = 1\{Y_{m,p}^{LLM} = t\}$ , where  $m \in \{\text{GPT-3.5, GPT-4o}\}$  denotes the LLM model and  $p \in \{1, \dots, 12\}$  denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

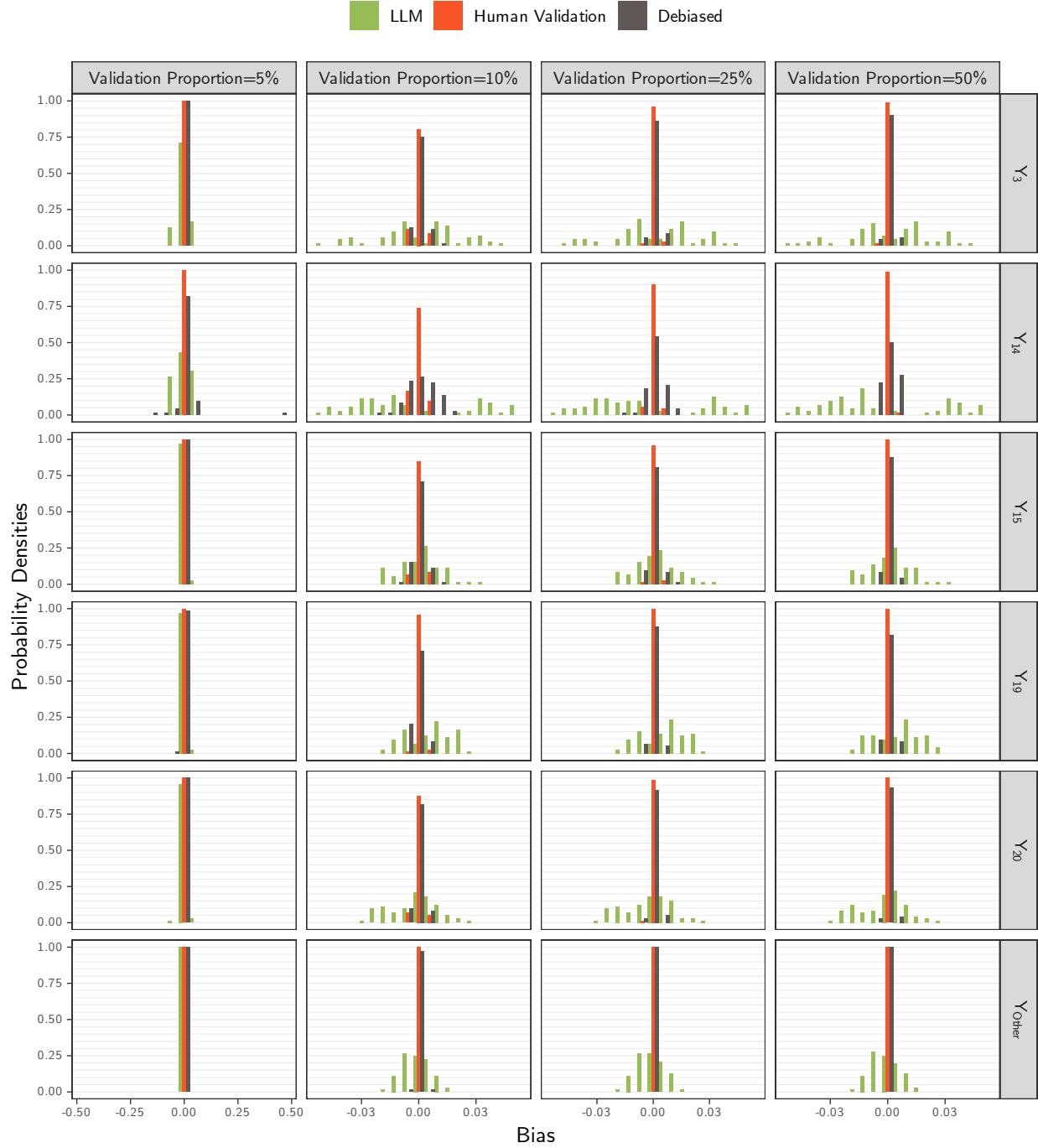
Figure 11: t-scores of each  $\beta_t$  vs. Prompt. Proxy on the RHS.



Note: The t-scores were obtained by fitting the model  $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$  over 10,000 bills, with 95% confidence intervals computed using robust standard errors. Here,  $Y_t$  denotes the indicator for topic  $t$ , where  $Y_t = 1\{Y = t\}$  for  $t \in \mathcal{T}$  and  $Y_{\text{Other}} = 1\{Y \notin \mathcal{T}\}$ , with  $\mathcal{T} = \{3, 14, 15, 19, 20\}$  is the 5 most-common human-labeled topics in the bills data. The variable  $V$  takes one of the following bills variables: Democrat, Senate, or DW1, where  $\text{extDemocrat} = 1\{\text{Party} = \text{Democrat}\}$ ,  $\text{Senate} = 1\{\text{Chamber} = \text{Senate}\}$ , and DW1 is the imputed DW1 score for the bill sponsor. The regressions based on human-labeled topics (red) were estimated with  $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$ , while the LLM-based regressions (GPT-3.5 in green, circles; and GPT-4o in blue, triangles) were estimated with  $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$ , where  $m \in \{\text{GPT-3.5}, \text{GPT-4o}\}$  denotes the LLM model and  $p \in \{1, \dots, 12\}$  denotes the prompt modifications. In each subplot, the indices for the prompts reflect the order chosen based on sorting the estimated values of the GPT-3.5 model in ascending order. The corresponding estimates for GPT-4o are aligned according to this order.

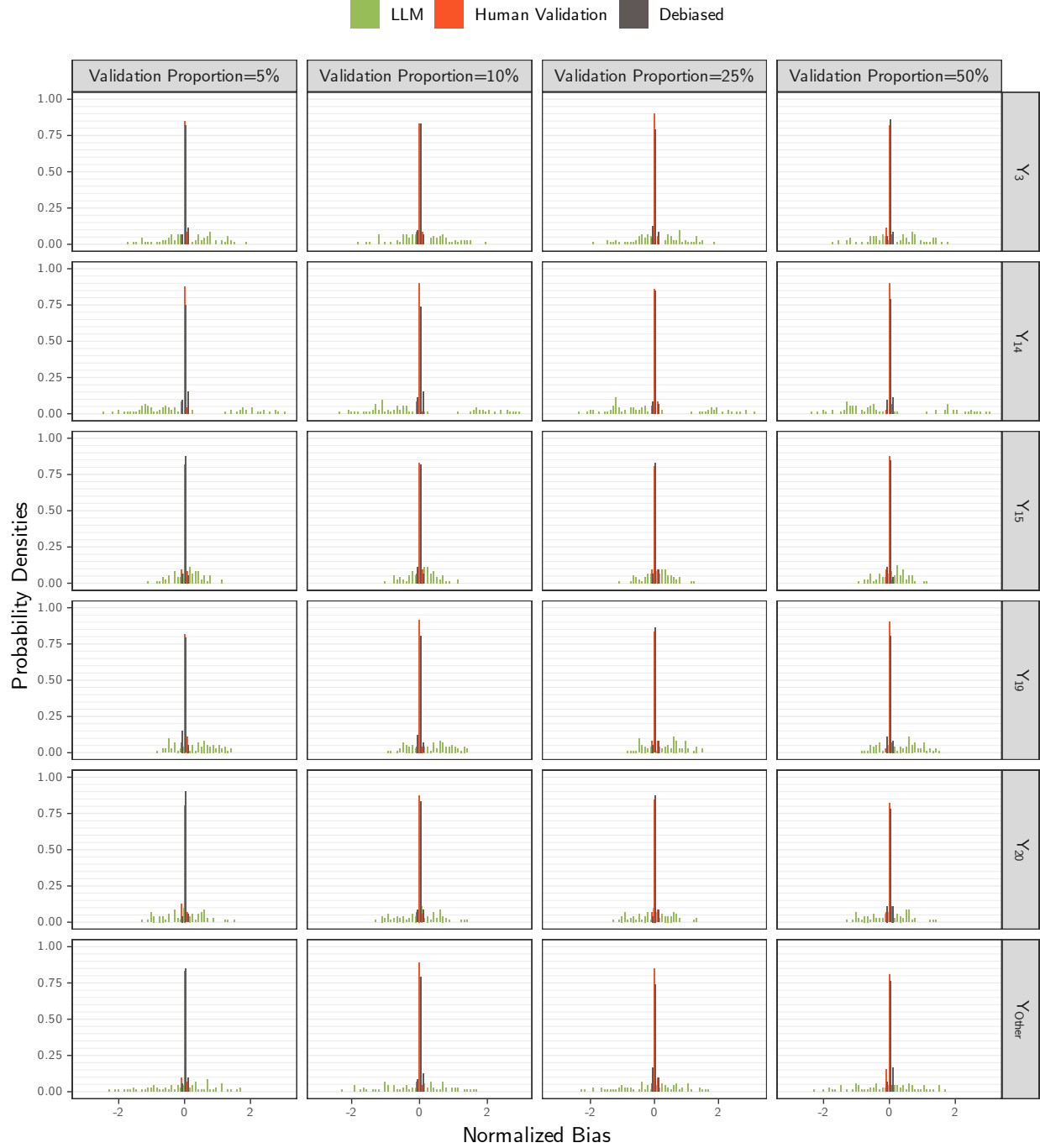


Figure 12: Bias Density Plots for all  $\beta$ s by Proportion of Validation Samples. Proxy on the RHS.



Note: The bias estimates were computed as the difference between the coefficient estimates from  $N = 1000$  simulation runs,  $\hat{\beta}_{V,\eta,m,p} = \sum_{i=1}^N \beta_{t,V,\eta,m,p}^{(i)} / N$ , and the human-based coefficient from the entire set of 10,000 bills,  $\beta_{t,V}^*$ , using a regression model of the form  $V = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$ , for topic  $t \in \{3, 14, 15, 19, 20, \text{Other}\}$ , variable  $V \in \{\text{Democrat, Senate, DW1}\}$ , validation proportion  $\eta \in \{5\%, 10\%, 25\%, 50\%\}$ , LLM model  $m \in \{\text{GPT-3.5, GPT-4o}\}$ , and prompt  $p \in \{1, \dots, 12\}$ . For the LLM-based models (green), a random sample of 5,000 bills drawn from the original 10,000 was used, with LLM-based labels,  $Y_{t,m,p}^{\text{LLM}} = 1\{Y_{m,p}^{\text{LLM}} = t\}$ . For the other two models—human-based (red) and debiased-based (brown)—the 5,000 sample were split into two subsets: validation and primary, with the size of the validation set determined by the validation proportion,  $\eta$ . In the human-based model, only the human labels in the validation set were used,  $Y_t^{\text{Human}} = 1\{Y^{\text{Human}} = t\}$ , whereas the debiased estimates use both the validation and primary sets, with  $Y_t^{\text{Human}}$  from validation set used to estimate measurement errors and  $Y_{t,m,p}^{\text{LLM}}$  from the primary set used to obtain the coefficient estimates used to compute the reported bias.

Figure 13: Normalized Bias Density of  $\beta_1$  by Proportion of Validation Samples. Proxy on the LHS.



Note: The normalized bias estimates were computed as the normalized difference between the coefficient estimates from  $N = 1000$  simulation runs,  $\bar{\beta}_{t,V,\eta,m,p}$ , and the human-based coefficient from the entire set of 10,000 bills,  $\beta_{t,V}^*$ , divided by the sample standard deviation,  $s_{t,V,\eta,m,p}$ , where  $s_{t,V,\eta,m,p}^2 = \sum_{i=1}^N (\beta_{t,V,\eta,m,p}^{(i)} - \bar{\beta}_{t,V,\eta,m,p})^2 / (N - 1)$  and all variables are as defined previously.

Figure 14: Bias Density of  $\beta_3$  by Model and Proportion of Validation Samples. Proxy on the RHS.

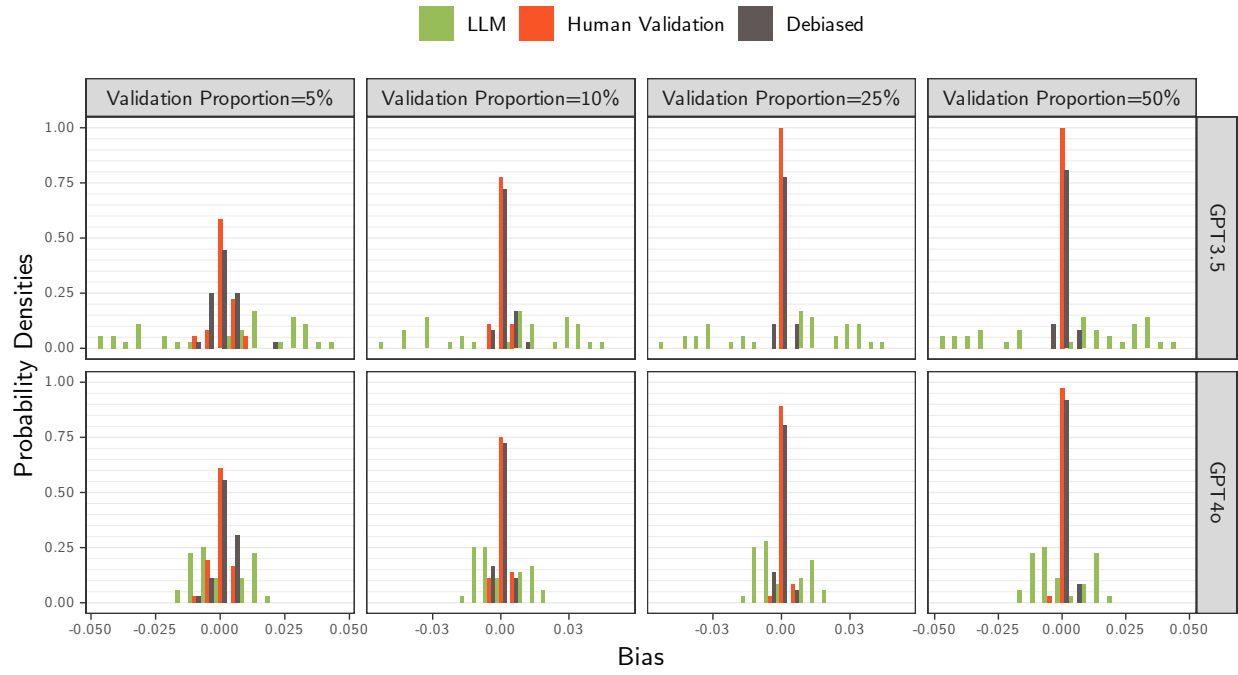


Figure 15: Bias Density of  $\beta_{14}$  by Model and Proportion of Validation Samples. Proxy on the RHS.

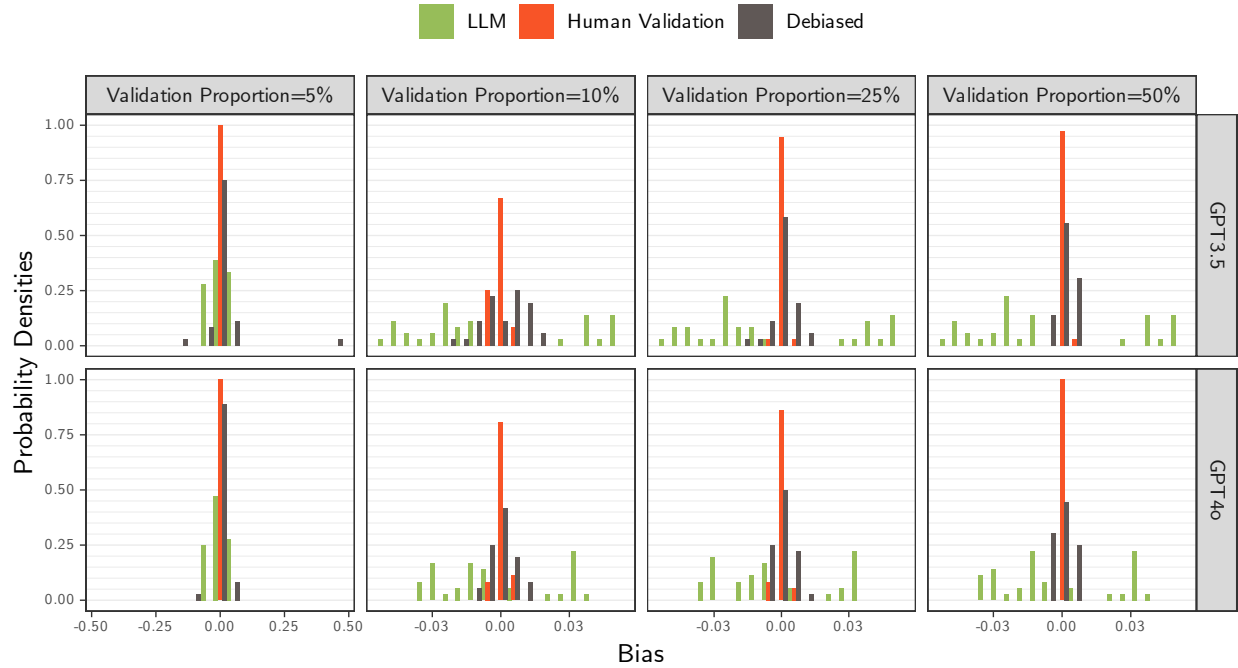


Figure 16: Bias Density of  $\beta_{15}$  by Model and Proportion of Validation Samples. Proxy on the RHS.

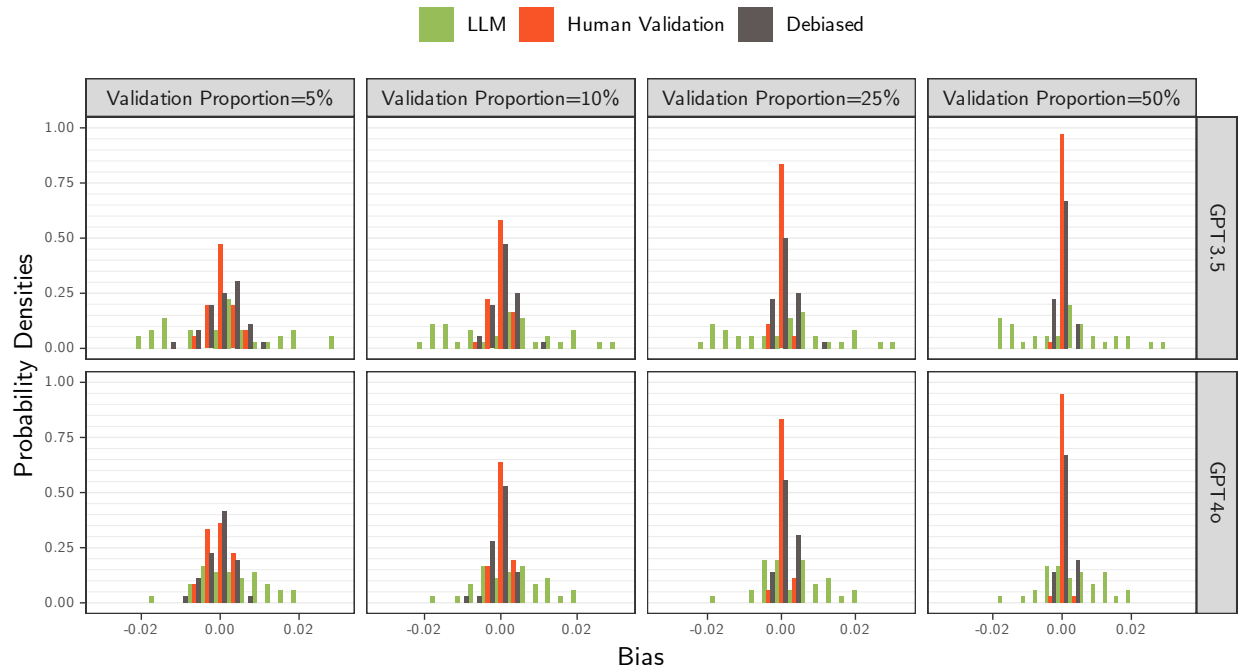


Figure 17: Bias Density of  $\beta_{19}$  by Model and Proportion of Validation Samples. Proxy on the RHS.

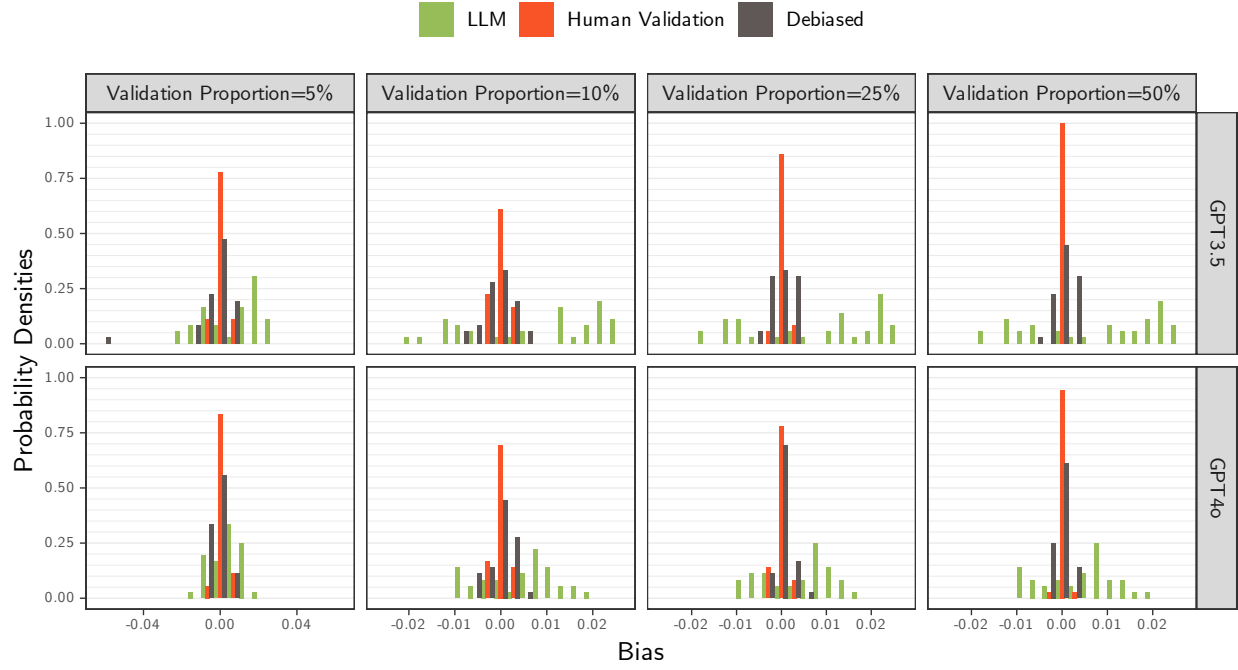


Figure 18: Bias Density of  $\beta_{20}$  by Model and Proportion of Validation Samples. Proxy on the RHS.



Figure 19: Bias Density of  $\beta_{\text{Other}}$  by Model and Proportion of Validation Samples. Proxy on the RHS.

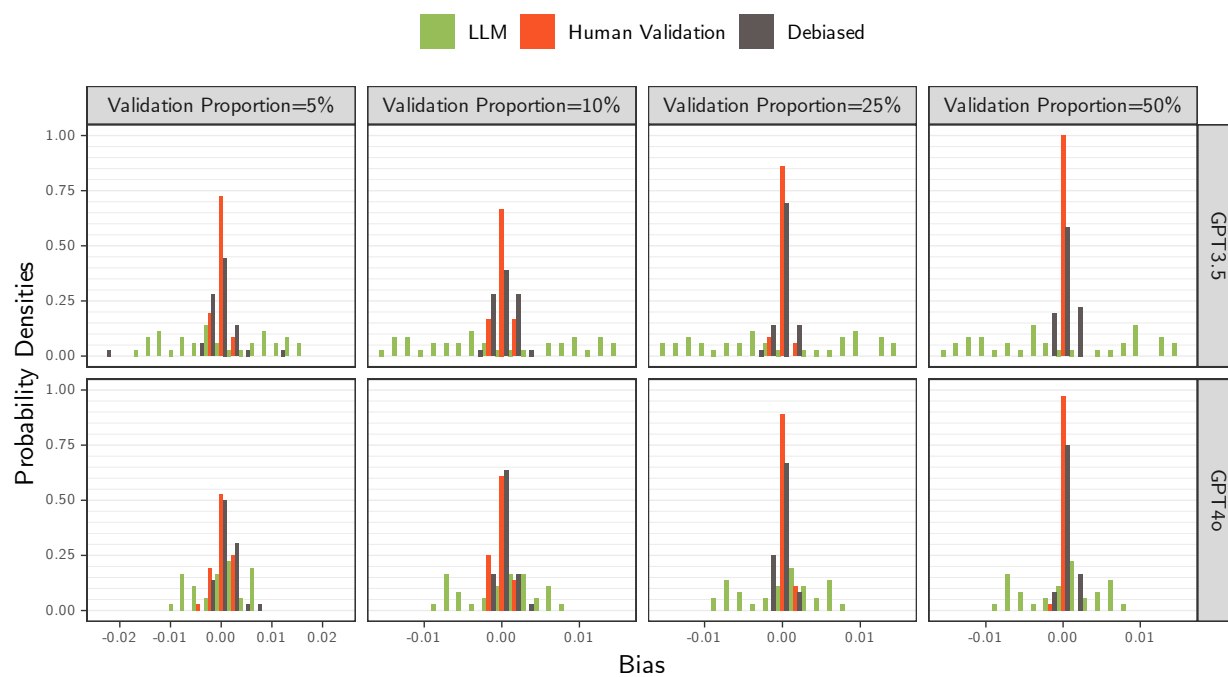


Figure 20: Normalized Bias Density of  $\beta_3$  by Model and Proportion of Validation Samples. Proxy on the RHS.

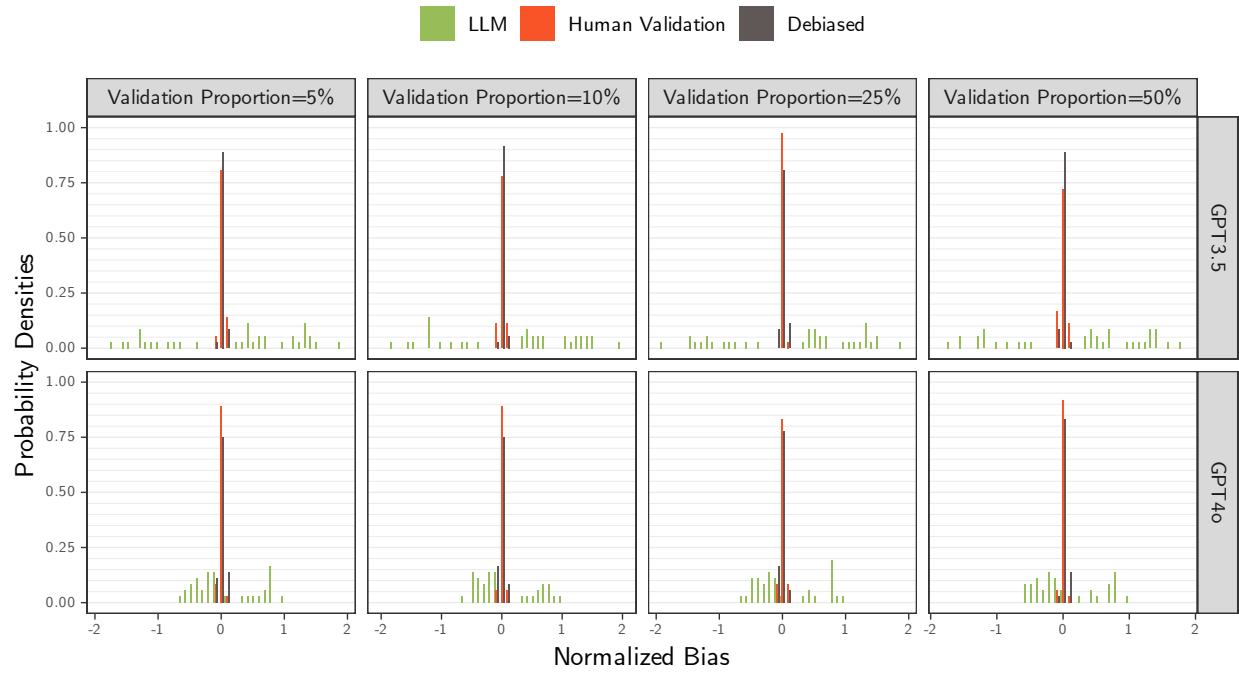


Figure 21: Normalized Bias Density of  $\beta_{14}$  by Model and Proportion of Validation Samples. Proxy on the RHS.

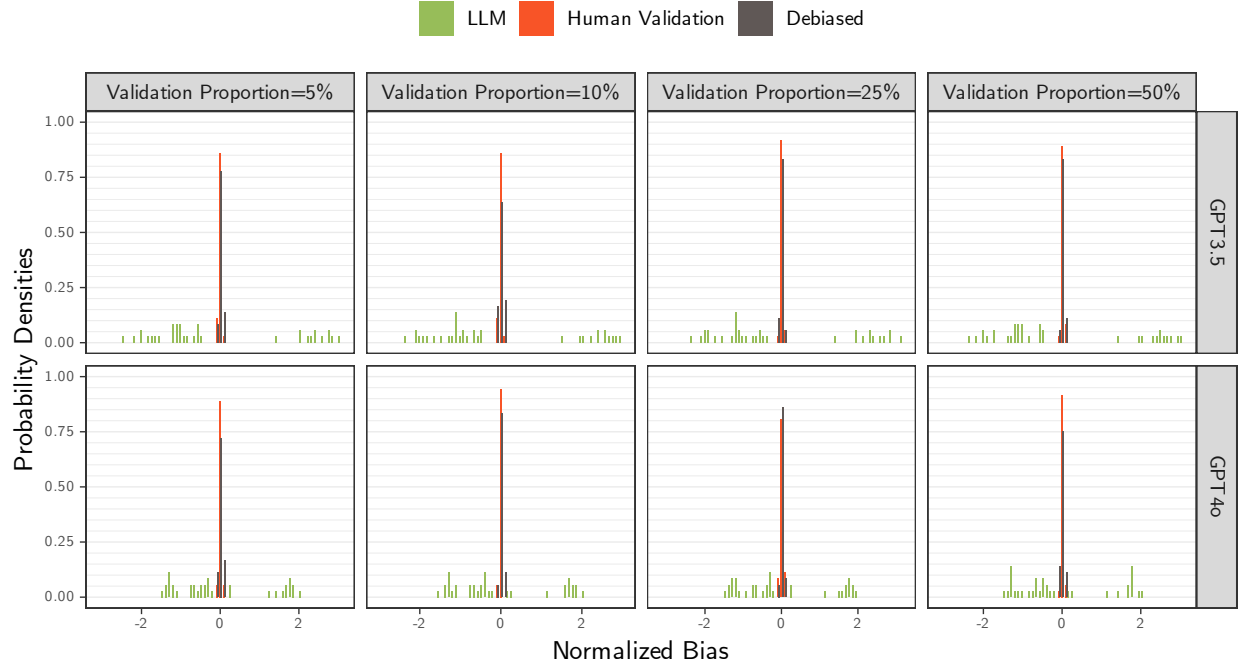


Figure 22: Normalized Bias Density of  $\beta_{15}$  by Model and Proportion of Validation Samples. Proxy on the RHS.

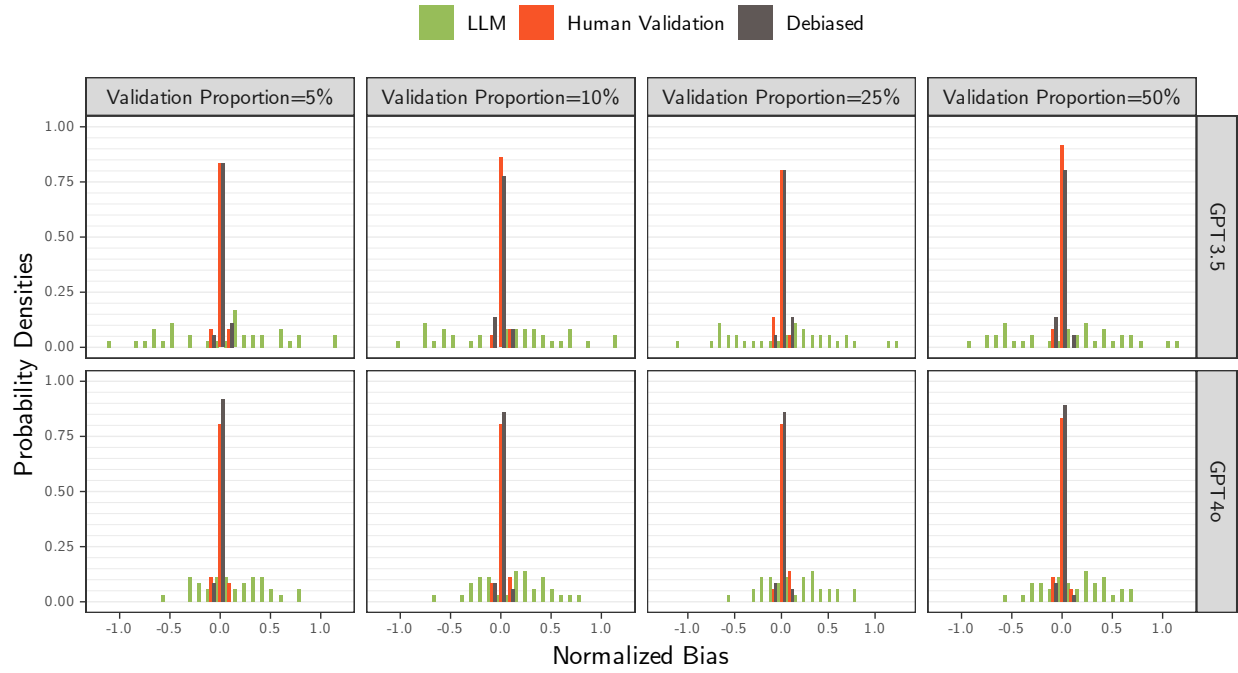




Figure 23: Normalized Bias Density of  $\beta_{19}$  by Model and Proportion of Validation Samples. Proxy on the RHS.



Figure 24: Normalized Bias Density of  $\beta_{20}$  by Model and Proportion of Validation Samples. Proxy on the RHS.



Figure 25: Normalized Bias Density of  $\beta_{\text{Other}}$  by Model and Proportion of Validation Samples. Proxy on the RHS.

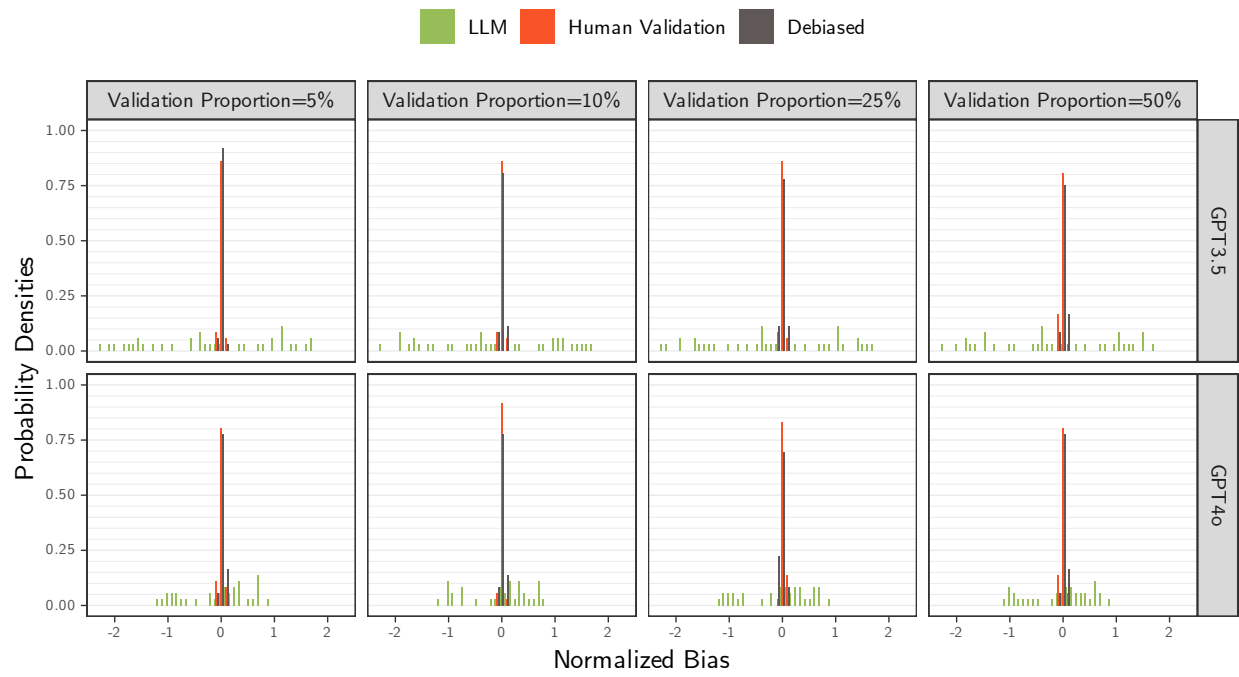


Figure 26: MSE Empirical CDF for all  $\beta$ s by Proportion of Validation Samples. Proxy on the RHS.

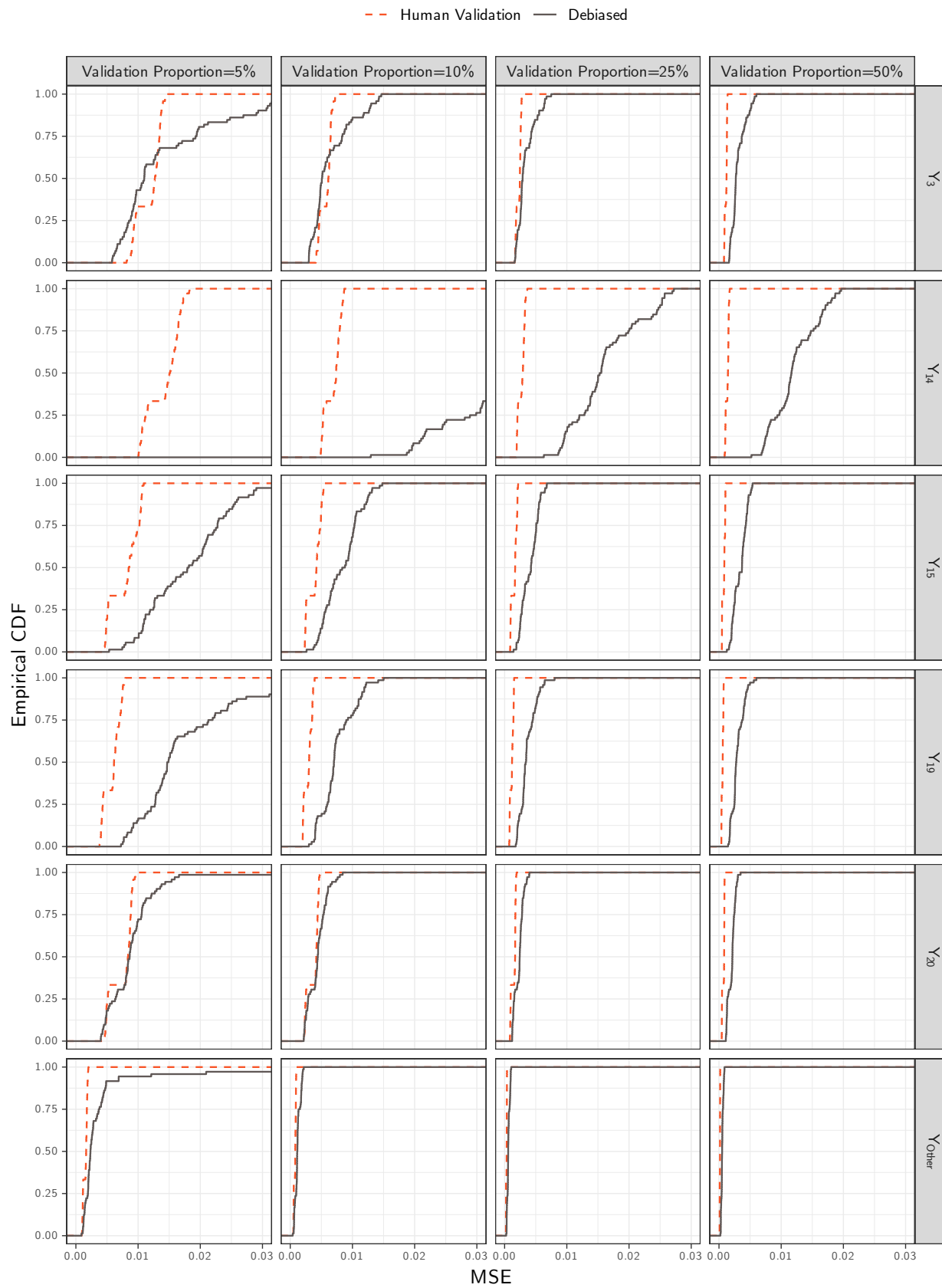


Figure 27: MSE Empirical CDF of  $\beta_3$  by Model and Proportion of Validation Samples. Proxy on the RHS.

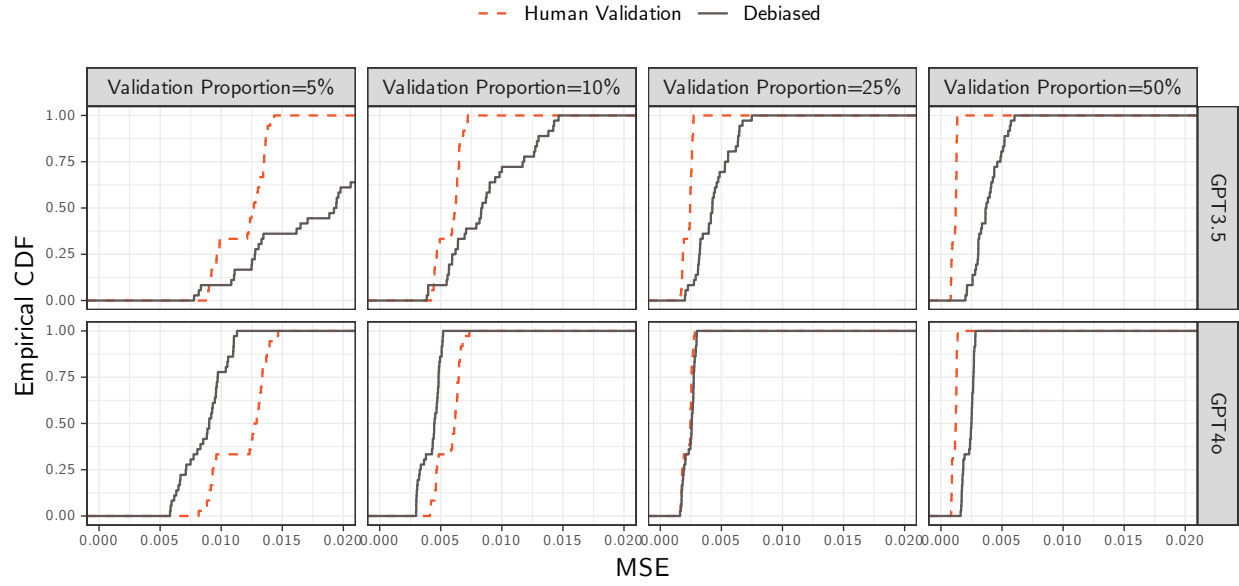


Figure 28: MSE Empirical CDF of  $\beta_{14}$  by Model and Proportion of Validation Samples. Proxy on the RHS.

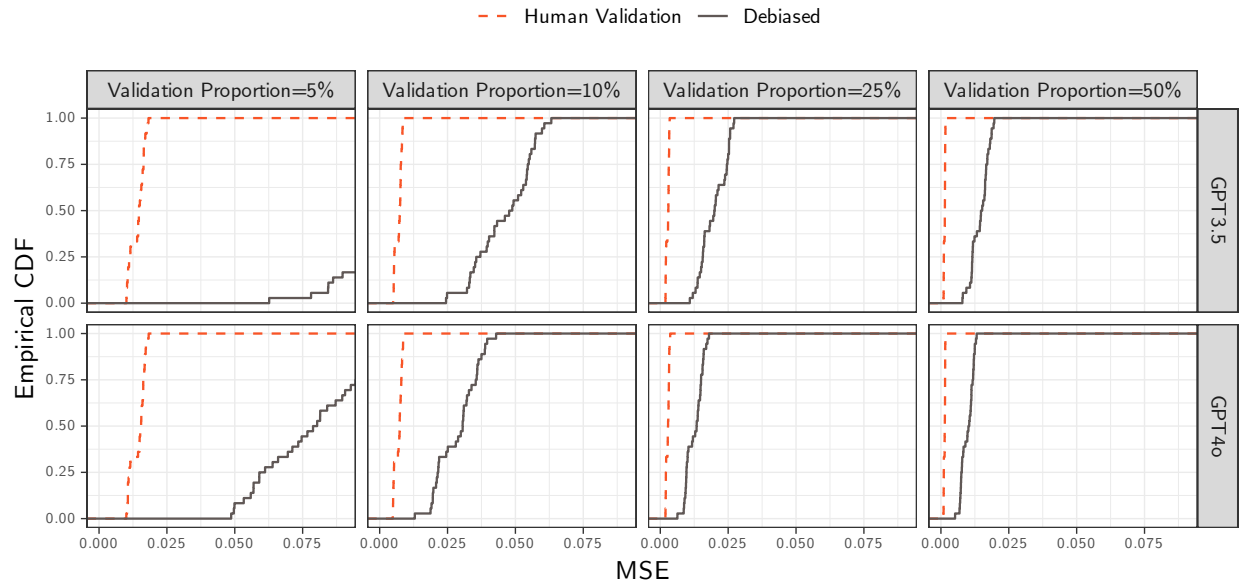


Figure 29: MSE Empirical CDF of  $\beta_{15}$  by Model and Proportion of Validation Samples. Proxy on the RHS.

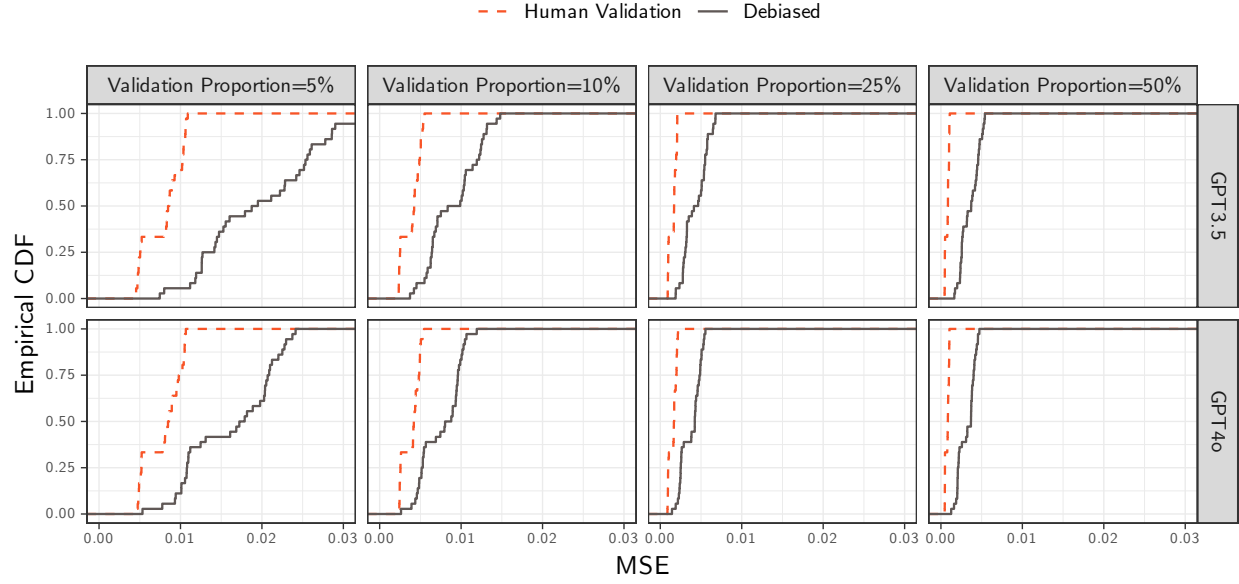


Figure 30: MSE Empirical CDF of  $\beta_{19}$  by Model and Proportion of Validation Samples. Proxy on the RHS.

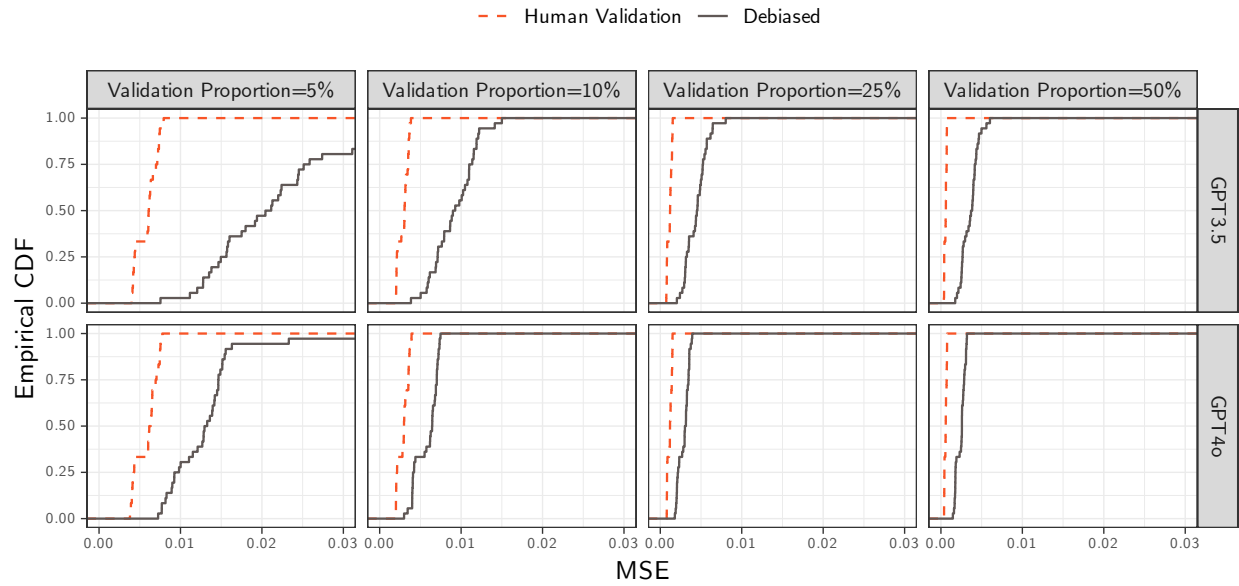


Figure 31: MSE Empirical CDF of  $\beta_{20}$  by Model and Proportion of Validation Samples. Proxy on the RHS.

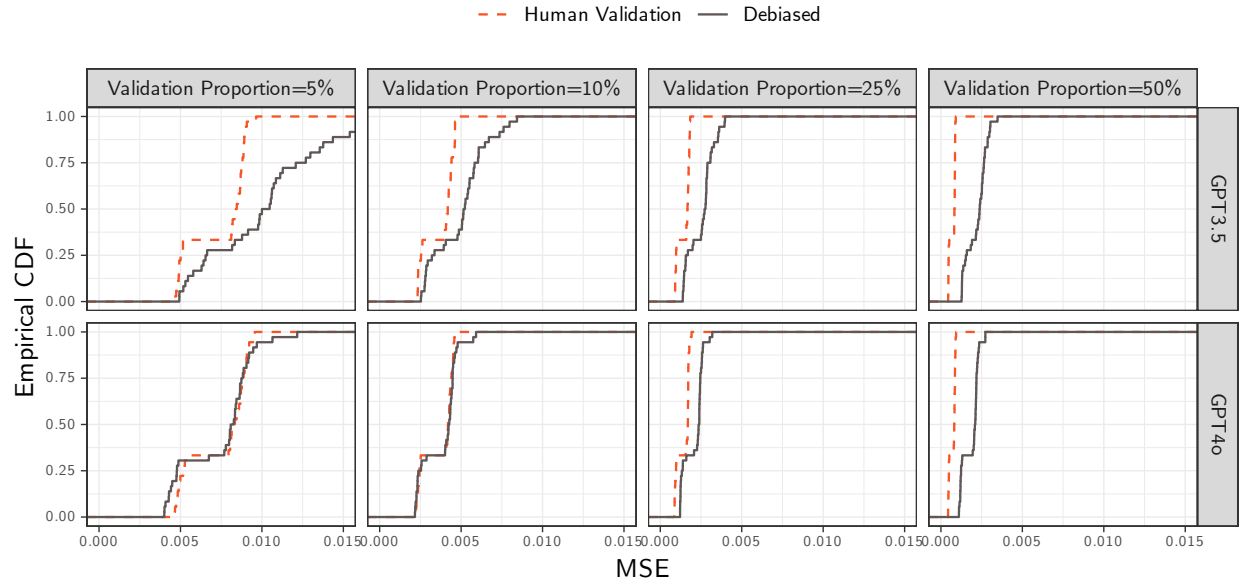


Figure 32: MSE Empirical CDF of  $\beta_{\text{Other}}$  by Model and Proportion of Validation Samples. Proxy on the RHS.

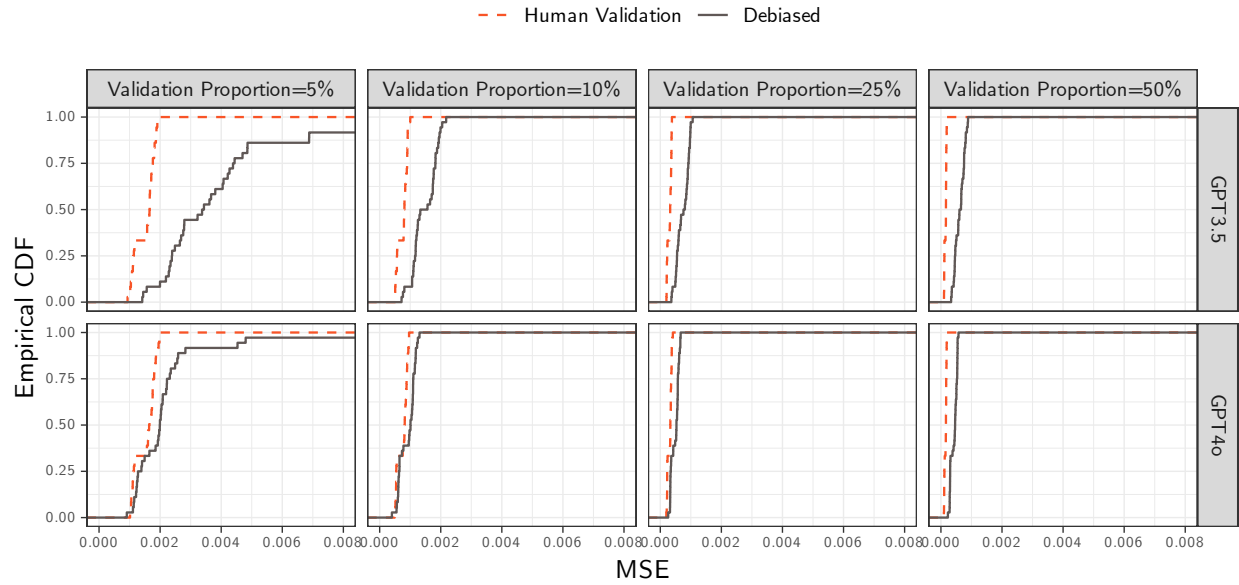


Table 9: Bias Summary Statistics of  $\beta_3$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Bias             | Mean  | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|-------|-------|--------|--------|-------|
| 5%                    | LLM              | 0.003 | 0.021 | 0.004  | -0.035 | 0.035 |
|                       | Human Validation | 0.000 | 0.003 | 0.001  | -0.005 | 0.006 |
|                       | Debiased         | 0.001 | 0.004 | 0.000  | -0.005 | 0.006 |
| 10%                   | LLM              | 0.003 | 0.022 | 0.003  | -0.035 | 0.035 |
|                       | Human Validation | 0.000 | 0.002 | 0.000  | -0.004 | 0.004 |
|                       | Debiased         | 0.000 | 0.003 | 0.000  | -0.004 | 0.004 |
| 25%                   | LLM              | 0.003 | 0.022 | 0.004  | -0.036 | 0.035 |
|                       | Human Validation | 0.000 | 0.001 | 0.000  | -0.002 | 0.002 |
|                       | Debiased         | 0.000 | 0.002 | 0.000  | -0.003 | 0.003 |
| 50%                   | LLM              | 0.003 | 0.022 | 0.003  | -0.036 | 0.036 |
|                       | Human Validation | 0.000 | 0.001 | 0.000  | -0.002 | 0.002 |
|                       | Debiased         | 0.000 | 0.002 | 0.000  | -0.002 | 0.003 |

Table 10: Bias Summary Statistics of  $\beta_{14}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Bias             | Mean   | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|--------|-------|--------|--------|-------|
| 5%                    | LLM              | -0.003 | 0.031 | -0.012 | -0.043 | 0.049 |
|                       | Human Validation | 0.000  | 0.003 | 0.000  | -0.006 | 0.005 |
|                       | Debiased         | 0.007  | 0.063 | 0.000  | -0.030 | 0.042 |
| 10%                   | LLM              | -0.003 | 0.031 | -0.012 | -0.043 | 0.050 |
|                       | Human Validation | 0.000  | 0.002 | -0.001 | -0.004 | 0.003 |
|                       | Debiased         | 0.001  | 0.008 | 0.001  | -0.010 | 0.013 |
| 25%                   | LLM              | -0.003 | 0.031 | -0.012 | -0.043 | 0.050 |
|                       | Human Validation | 0.000  | 0.002 | 0.000  | -0.003 | 0.003 |
|                       | Debiased         | 0.000  | 0.004 | 0.001  | -0.007 | 0.006 |
| 50%                   | LLM              | -0.003 | 0.031 | -0.012 | -0.043 | 0.048 |
|                       | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.002 |
|                       | Debiased         | 0.000  | 0.004 | 0.000  | -0.005 | 0.006 |

Table 11: Bias Summary Statistics of  $\beta_{15}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Bias             | Mean  | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|-------|-------|--------|--------|-------|
| 5%                    | LLM              | 0.003 | 0.011 | 0.003  | -0.017 | 0.020 |
|                       | Human Validation | 0.000 | 0.003 | 0.000  | -0.005 | 0.004 |
|                       | Debiased         | 0.000 | 0.004 | 0.000  | -0.007 | 0.006 |
| 10%                   | LLM              | 0.002 | 0.011 | 0.003  | -0.018 | 0.020 |
|                       | Human Validation | 0.000 | 0.002 | 0.000  | -0.003 | 0.003 |
|                       | Debiased         | 0.000 | 0.003 | 0.000  | -0.005 | 0.004 |
| 25%                   | LLM              | 0.003 | 0.011 | 0.003  | -0.017 | 0.020 |
|                       | Human Validation | 0.000 | 0.001 | 0.000  | -0.002 | 0.002 |
|                       | Debiased         | 0.000 | 0.002 | 0.001  | -0.003 | 0.003 |
| 50%                   | LLM              | 0.003 | 0.011 | 0.003  | -0.017 | 0.019 |
|                       | Human Validation | 0.000 | 0.001 | 0.000  | -0.002 | 0.001 |
|                       | Debiased         | 0.000 | 0.002 | 0.000  | -0.003 | 0.003 |

Table 12: Bias Summary Statistics of  $\beta_{19}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Bias             | Mean   | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|--------|-------|--------|--------|-------|
| 5%                    | LLM              | 0.007  | 0.012 | 0.009  | -0.011 | 0.023 |
|                       | Human Validation | 0.001  | 0.003 | 0.001  | -0.004 | 0.005 |
|                       | Debiased         | -0.002 | 0.009 | -0.001 | -0.010 | 0.005 |
| 10%                   | LLM              | 0.007  | 0.012 | 0.009  | -0.011 | 0.024 |
|                       | Human Validation | 0.000  | 0.002 | 0.000  | -0.003 | 0.002 |
|                       | Debiased         | 0.000  | 0.003 | 0.000  | -0.005 | 0.004 |
| 25%                   | LLM              | 0.006  | 0.012 | 0.009  | -0.011 | 0.024 |
|                       | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.002 |
|                       | Debiased         | 0.000  | 0.002 | 0.000  | -0.003 | 0.003 |
| 50%                   | LLM              | 0.006  | 0.012 | 0.009  | -0.011 | 0.024 |
|                       | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.001 |
|                       | Debiased         | 0.000  | 0.002 | 0.000  | -0.003 | 0.003 |

Table 13: Bias Summary Statistics of  $\beta_{20}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Bias             | Mean   | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|--------|-------|--------|--------|-------|
| 5%                    | LLM              | -0.001 | 0.013 | 0.001  | -0.022 | 0.016 |
|                       | Human Validation | 0.000  | 0.003 | 0.000  | -0.005 | 0.004 |
|                       | Debiased         | 0.000  | 0.003 | 0.000  | -0.004 | 0.005 |
| 10%                   | LLM              | -0.001 | 0.013 | 0.001  | -0.023 | 0.015 |
|                       | Human Validation | 0.000  | 0.002 | 0.000  | -0.003 | 0.003 |
|                       | Debiased         | 0.000  | 0.002 | 0.000  | -0.003 | 0.004 |
| 25%                   | LLM              | -0.001 | 0.013 | 0.001  | -0.021 | 0.015 |
|                       | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.002 |
|                       | Debiased         | 0.000  | 0.001 | 0.000  | -0.002 | 0.003 |
| 50%                   | LLM              | -0.001 | 0.012 | 0.001  | -0.021 | 0.015 |
|                       | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.002 |
|                       | Debiased         | 0.000  | 0.002 | 0.000  | -0.003 | 0.003 |

Table 14: Bias Summary Statistics of  $\beta_{\text{Other}}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Bias             | Mean | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|------|-------|--------|--------|-------|
| 5%                    | LLM              | 0    | 0.008 | 0      | -0.012 | 0.013 |
|                       | Human Validation | 0    | 0.001 | 0      | -0.002 | 0.002 |
|                       | Debiased         | 0    | 0.003 | 0      | -0.003 | 0.003 |
| 10%                   | LLM              | 0    | 0.008 | 0      | -0.012 | 0.013 |
|                       | Human Validation | 0    | 0.001 | 0      | -0.001 | 0.001 |
|                       | Debiased         | 0    | 0.001 | 0      | -0.002 | 0.002 |
| 25%                   | LLM              | 0    | 0.008 | 0      | -0.012 | 0.014 |
|                       | Human Validation | 0    | 0.001 | 0      | -0.001 | 0.001 |
|                       | Debiased         | 0    | 0.001 | 0      | -0.002 | 0.001 |
| 50%                   | LLM              | 0    | 0.008 | 0      | -0.012 | 0.013 |
|                       | Human Validation | 0    | 0.000 | 0      | -0.001 | 0.001 |
|                       | Debiased         | 0    | 0.001 | 0      | -0.001 | 0.001 |



Table 15: Normalized Bias Summary Statistics of  $\beta_3$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|--------|-------|--------|--------|-------|
| 5%                    | LLM              | 0.148  | 0.846 | 0.166  | -1.236 | 1.398 |
|                       | Human Validation | 0.001  | 0.032 | 0.005  | -0.052 | 0.057 |
|                       | Debiased         | 0.003  | 0.033 | 0.003  | -0.051 | 0.051 |
| 10%                   | LLM              | 0.150  | 0.842 | 0.139  | -1.202 | 1.419 |
|                       | Human Validation | -0.002 | 0.029 | -0.006 | -0.051 | 0.049 |
|                       | Debiased         | 0.000  | 0.033 | -0.003 | -0.056 | 0.050 |
| 25%                   | LLM              | 0.154  | 0.852 | 0.165  | -1.287 | 1.419 |
|                       | Human Validation | 0.006  | 0.028 | 0.008  | -0.040 | 0.047 |
|                       | Debiased         | -0.001 | 0.035 | -0.001 | -0.055 | 0.057 |
| 50%                   | LLM              | 0.150  | 0.845 | 0.124  | -1.286 | 1.444 |
|                       | Human Validation | 0.001  | 0.032 | 0.003  | -0.053 | 0.045 |
|                       | Debiased         | -0.002 | 0.029 | -0.004 | -0.046 | 0.047 |

Table 16: Normalized Bias Summary Statistics of  $\beta_{14}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|--------|-------|--------|--------|-------|
| 5%                    | LLM              | 0.013  | 1.541 | -0.537 | -1.885 | 2.673 |
|                       | Human Validation | -0.002 | 0.029 | -0.001 | -0.050 | 0.044 |
|                       | Debiased         | 0.004  | 0.039 | 0.002  | -0.057 | 0.063 |
| 10%                   | LLM              | 0.013  | 1.543 | -0.553 | -1.929 | 2.667 |
|                       | Human Validation | -0.006 | 0.028 | -0.006 | -0.046 | 0.043 |
|                       | Debiased         | 0.002  | 0.039 | 0.003  | -0.058 | 0.067 |
| 25%                   | LLM              | 0.005  | 1.541 | -0.560 | -1.934 | 2.622 |
|                       | Human Validation | 0.001  | 0.032 | -0.001 | -0.047 | 0.050 |
|                       | Debiased         | -0.001 | 0.034 | 0.005  | -0.059 | 0.056 |
| 50%                   | LLM              | 0.009  | 1.547 | -0.543 | -1.968 | 2.667 |
|                       | Human Validation | 0.003  | 0.032 | 0.004  | -0.042 | 0.050 |
|                       | Debiased         | 0.001  | 0.034 | 0.001  | -0.057 | 0.057 |

Table 17: Normalized Bias Summary Statistics of  $\beta_{15}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|--------|-------|--------|--------|-------|
| 5%                    | LLM              | 0.118  | 0.463 | 0.166  | -0.633 | 0.791 |
|                       | Human Validation | -0.003 | 0.033 | -0.004 | -0.056 | 0.050 |
|                       | Debiased         | -0.002 | 0.030 | -0.002 | -0.046 | 0.045 |
| 10%                   | LLM              | 0.108  | 0.470 | 0.167  | -0.695 | 0.764 |
|                       | Human Validation | -0.001 | 0.034 | 0.000  | -0.047 | 0.056 |
|                       | Debiased         | -0.003 | 0.033 | -0.002 | -0.060 | 0.048 |
| 25%                   | LLM              | 0.121  | 0.460 | 0.150  | -0.653 | 0.782 |
|                       | Human Validation | -0.005 | 0.034 | -0.012 | -0.055 | 0.056 |
|                       | Debiased         | 0.004  | 0.035 | 0.007  | -0.052 | 0.054 |
| 50%                   | LLM              | 0.112  | 0.450 | 0.158  | -0.649 | 0.762 |
|                       | Human Validation | -0.006 | 0.028 | -0.003 | -0.052 | 0.035 |
|                       | Debiased         | -0.002 | 0.030 | 0.000  | -0.051 | 0.042 |

Table 18: Normalized Bias Summary Statistics of  $\beta_{19}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|--------|-------|--------|--------|-------|
| 5%                    | LLM              | 0.369  | 0.615 | 0.486  | -0.534 | 1.285 |
|                       | Human Validation | 0.008  | 0.034 | 0.010  | -0.050 | 0.062 |
|                       | Debiased         | -0.010 | 0.034 | -0.009 | -0.064 | 0.039 |
| 10%                   | LLM              | 0.368  | 0.615 | 0.466  | -0.502 | 1.344 |
|                       | Human Validation | -0.004 | 0.029 | -0.006 | -0.044 | 0.039 |
|                       | Debiased         | -0.004 | 0.034 | -0.003 | -0.064 | 0.048 |
| 25%                   | LLM              | 0.359  | 0.607 | 0.451  | -0.493 | 1.326 |
|                       | Human Validation | -0.001 | 0.032 | -0.001 | -0.058 | 0.055 |
|                       | Debiased         | 0.000  | 0.033 | -0.002 | -0.045 | 0.054 |
| 50%                   | LLM              | 0.361  | 0.619 | 0.466  | -0.554 | 1.331 |
|                       | Human Validation | 0.007  | 0.027 | 0.007  | -0.034 | 0.047 |
|                       | Debiased         | -0.002 | 0.035 | 0.000  | -0.069 | 0.049 |

Table 19: Normalized Bias Summary Statistics of  $\beta_{20}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|--------|-------|--------|--------|-------|
| 5%                    | LLM              | -0.008 | 0.632 | 0.023  | -1.000 | 0.888 |
|                       | Human Validation | -0.001 | 0.034 | 0.001  | -0.059 | 0.047 |
|                       | Debiased         | 0.002  | 0.030 | 0.001  | -0.041 | 0.048 |
| 10%                   | LLM              | -0.002 | 0.630 | 0.041  | -1.005 | 0.864 |
|                       | Human Validation | 0.001  | 0.030 | 0.002  | -0.049 | 0.043 |
|                       | Debiased         | 0.004  | 0.033 | 0.004  | -0.048 | 0.056 |
| 25%                   | LLM              | -0.009 | 0.623 | 0.029  | -0.970 | 0.838 |
|                       | Human Validation | 0.003  | 0.030 | 0.004  | -0.049 | 0.047 |
|                       | Debiased         | -0.001 | 0.030 | -0.007 | -0.040 | 0.060 |
| 50%                   | LLM              | -0.003 | 0.619 | 0.049  | -0.990 | 0.835 |
|                       | Human Validation | 0.001  | 0.034 | 0.000  | -0.055 | 0.054 |
|                       | Debiased         | 0.000  | 0.036 | 0.003  | -0.055 | 0.061 |

Table 20: Normalized Bias Summary Statistics of  $\beta_{\text{Other}}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|-----------------------|------------------|--------|-------|--------|--------|-------|
| 5%                    | LLM              | -0.093 | 0.979 | 0.030  | -1.766 | 1.413 |
|                       | Human Validation | -0.003 | 0.033 | -0.006 | -0.051 | 0.055 |
|                       | Debiased         | 0.001  | 0.031 | -0.006 | -0.046 | 0.055 |
| 10%                   | LLM              | -0.100 | 0.961 | 0.021  | -1.788 | 1.394 |
|                       | Human Validation | -0.001 | 0.030 | 0.001  | -0.049 | 0.044 |
|                       | Debiased         | 0.000  | 0.035 | -0.004 | -0.049 | 0.060 |
| 25%                   | LLM              | -0.100 | 0.970 | 0.016  | -1.767 | 1.439 |
|                       | Human Validation | 0.001  | 0.030 | -0.002 | -0.045 | 0.053 |
|                       | Debiased         | -0.005 | 0.036 | -0.004 | -0.065 | 0.055 |
| 50%                   | LLM              | -0.082 | 0.957 | 0.036  | -1.721 | 1.424 |
|                       | Human Validation | -0.001 | 0.034 | 0.008  | -0.057 | 0.043 |
|                       | Debiased         | 0.004  | 0.037 | 0.001  | -0.050 | 0.060 |

Table 21: Bias Summary Statistics of  $\beta_3$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | Bias             | Mean   | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|--------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.004  | 0.028 | 0.013  | -0.042 | 0.037 |
| GPT-3.5 | 5%                    | Human Validation | 0.001  | 0.004 | 0.001  | -0.005 | 0.006 |
| GPT-3.5 | 5%                    | Debiased         | 0.001  | 0.005 | 0.000  | -0.005 | 0.007 |
| GPT-3.5 | 10%                   | LLM              | 0.004  | 0.029 | 0.012  | -0.041 | 0.038 |
| GPT-3.5 | 10%                   | Human Validation | 0.000  | 0.002 | 0.000  | -0.004 | 0.004 |
| GPT-3.5 | 10%                   | Debiased         | 0.000  | 0.003 | 0.000  | -0.003 | 0.004 |
| GPT-3.5 | 25%                   | LLM              | 0.004  | 0.029 | 0.012  | -0.041 | 0.038 |
| GPT-3.5 | 25%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.002 |
| GPT-3.5 | 25%                   | Debiased         | 0.000  | 0.002 | 0.000  | -0.003 | 0.004 |
| GPT-3.5 | 50%                   | LLM              | 0.004  | 0.029 | 0.012  | -0.041 | 0.037 |
| GPT-3.5 | 50%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.002 |
| GPT-3.5 | 50%                   | Debiased         | 0.000  | 0.002 | 0.000  | -0.003 | 0.003 |
| GPT-4o  | 5%                    | LLM              | 0.001  | 0.011 | -0.003 | -0.012 | 0.017 |
| GPT-4o  | 5%                    | Human Validation | -0.001 | 0.003 | 0.000  | -0.005 | 0.004 |
| GPT-4o  | 5%                    | Debiased         | 0.000  | 0.003 | 0.000  | -0.005 | 0.005 |
| GPT-4o  | 10%                   | LLM              | 0.001  | 0.011 | -0.003 | -0.012 | 0.017 |
| GPT-4o  | 10%                   | Human Validation | 0.000  | 0.002 | 0.000  | -0.003 | 0.003 |
| GPT-4o  | 10%                   | Debiased         | 0.000  | 0.002 | 0.000  | -0.004 | 0.004 |
| GPT-4o  | 25%                   | LLM              | 0.001  | 0.011 | -0.003 | -0.012 | 0.017 |
| GPT-4o  | 25%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.003 |
| GPT-4o  | 25%                   | Debiased         | 0.000  | 0.002 | 0.000  | -0.003 | 0.003 |
| GPT-4o  | 50%                   | LLM              | 0.001  | 0.011 | -0.003 | -0.012 | 0.016 |
| GPT-4o  | 50%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-4o  | 50%                   | Debiased         | 0.000  | 0.002 | 0.000  | -0.002 | 0.003 |

Table 22: Bias Summary Statistics of  $\beta_{14}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | Bias             | Mean   | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|--------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | -0.004 | 0.036 | -0.018 | -0.044 | 0.051 |
| GPT-3.5 | 5%                    | Human Validation | -0.001 | 0.004 | -0.001 | -0.006 | 0.004 |
| GPT-3.5 | 5%                    | Debiased         | 0.012  | 0.086 | 0.000  | -0.034 | 0.051 |
| GPT-3.5 | 10%                   | LLM              | -0.003 | 0.036 | -0.019 | -0.045 | 0.052 |
| GPT-3.5 | 10%                   | Human Validation | -0.001 | 0.002 | -0.001 | -0.004 | 0.003 |
| GPT-3.5 | 10%                   | Debiased         | 0.001  | 0.009 | 0.002  | -0.012 | 0.013 |
| GPT-3.5 | 25%                   | LLM              | -0.004 | 0.036 | -0.017 | -0.046 | 0.052 |
| GPT-3.5 | 25%                   | Human Validation | 0.000  | 0.002 | 0.000  | -0.002 | 0.002 |
| GPT-3.5 | 25%                   | Debiased         | 0.000  | 0.005 | 0.001  | -0.008 | 0.006 |
| GPT-3.5 | 50%                   | LLM              | -0.004 | 0.036 | -0.020 | -0.045 | 0.051 |
| GPT-3.5 | 50%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.002 |
| GPT-3.5 | 50%                   | Debiased         | 0.001  | 0.004 | 0.000  | -0.005 | 0.006 |
| GPT-4o  | 5%                    | LLM              | -0.002 | 0.025 | -0.008 | -0.031 | 0.036 |
| GPT-4o  | 5%                    | Human Validation | 0.000  | 0.003 | 0.000  | -0.005 | 0.005 |
| GPT-4o  | 5%                    | Debiased         | 0.002  | 0.021 | 0.001  | -0.021 | 0.032 |
| GPT-4o  | 10%                   | LLM              | -0.002 | 0.025 | -0.009 | -0.032 | 0.036 |
| GPT-4o  | 10%                   | Human Validation | 0.000  | 0.002 | 0.000  | -0.004 | 0.003 |
| GPT-4o  | 10%                   | Debiased         | 0.000  | 0.006 | 0.000  | -0.009 | 0.012 |
| GPT-4o  | 25%                   | LLM              | -0.002 | 0.025 | -0.008 | -0.032 | 0.036 |
| GPT-4o  | 25%                   | Human Validation | 0.000  | 0.002 | 0.000  | -0.003 | 0.003 |
| GPT-4o  | 25%                   | Debiased         | 0.000  | 0.004 | 0.000  | -0.005 | 0.006 |
| GPT-4o  | 50%                   | LLM              | -0.002 | 0.025 | -0.009 | -0.032 | 0.035 |
| GPT-4o  | 50%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.002 |
| GPT-4o  | 50%                   | Debiased         | 0.000  | 0.004 | 0.000  | -0.005 | 0.005 |

Table 23: Bias Summary Statistics of  $\beta_{15}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | Bias             | Mean   | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|--------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.002  | 0.013 | 0.003  | -0.018 | 0.023 |
| GPT-3.5 | 5%                    | Human Validation | 0.000  | 0.003 | -0.001 | -0.004 | 0.005 |
| GPT-3.5 | 5%                    | Debiased         | 0.000  | 0.005 | 0.001  | -0.006 | 0.007 |
| GPT-3.5 | 10%                   | LLM              | 0.001  | 0.014 | 0.003  | -0.018 | 0.022 |
| GPT-3.5 | 10%                   | Human Validation | 0.000  | 0.002 | 0.000  | -0.003 | 0.003 |
| GPT-3.5 | 10%                   | Debiased         | 0.000  | 0.003 | 0.000  | -0.005 | 0.004 |
| GPT-3.5 | 25%                   | LLM              | 0.002  | 0.014 | 0.003  | -0.017 | 0.022 |
| GPT-3.5 | 25%                   | Human Validation | 0.000  | 0.001 | -0.001 | -0.002 | 0.002 |
| GPT-3.5 | 25%                   | Debiased         | 0.000  | 0.003 | 0.000  | -0.003 | 0.004 |
| GPT-3.5 | 50%                   | LLM              | 0.001  | 0.013 | 0.003  | -0.017 | 0.022 |
| GPT-3.5 | 50%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.001 |
| GPT-3.5 | 50%                   | Debiased         | 0.000  | 0.002 | 0.000  | -0.003 | 0.003 |
| GPT-4o  | 5%                    | LLM              | 0.004  | 0.008 | 0.003  | -0.008 | 0.016 |
| GPT-4o  | 5%                    | Human Validation | -0.001 | 0.003 | 0.000  | -0.005 | 0.004 |
| GPT-4o  | 5%                    | Debiased         | -0.001 | 0.004 | -0.001 | -0.007 | 0.004 |
| GPT-4o  | 10%                   | LLM              | 0.003  | 0.008 | 0.004  | -0.008 | 0.016 |
| GPT-4o  | 10%                   | Human Validation | 0.000  | 0.002 | 0.000  | -0.003 | 0.004 |
| GPT-4o  | 10%                   | Debiased         | -0.001 | 0.003 | 0.000  | -0.004 | 0.003 |
| GPT-4o  | 25%                   | LLM              | 0.004  | 0.008 | 0.003  | -0.007 | 0.017 |
| GPT-4o  | 25%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.003 |
| GPT-4o  | 25%                   | Debiased         | 0.000  | 0.002 | 0.001  | -0.004 | 0.003 |
| GPT-4o  | 50%                   | LLM              | 0.004  | 0.008 | 0.003  | -0.008 | 0.017 |
| GPT-4o  | 50%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.001 |
| GPT-4o  | 50%                   | Debiased         | 0.000  | 0.002 | 0.000  | -0.003 | 0.002 |

Table 24: Bias Summary Statistics of  $\beta_{19}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | Bias             | Mean   | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|--------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.008  | 0.015 | 0.013  | -0.013 | 0.024 |
| GPT-3.5 | 5%                    | Human Validation | 0.000  | 0.003 | 0.000  | -0.005 | 0.005 |
| GPT-3.5 | 5%                    | Debiased         | -0.003 | 0.012 | -0.001 | -0.012 | 0.005 |
| GPT-3.5 | 10%                   | LLM              | 0.009  | 0.015 | 0.014  | -0.013 | 0.024 |
| GPT-3.5 | 10%                   | Human Validation | 0.000  | 0.002 | 0.000  | -0.003 | 0.002 |
| GPT-3.5 | 10%                   | Debiased         | -0.001 | 0.003 | -0.001 | -0.006 | 0.004 |
| GPT-3.5 | 25%                   | LLM              | 0.008  | 0.015 | 0.014  | -0.013 | 0.025 |
| GPT-3.5 | 25%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.002 |
| GPT-3.5 | 25%                   | Debiased         | 0.000  | 0.002 | 0.000  | -0.004 | 0.003 |
| GPT-3.5 | 50%                   | LLM              | 0.008  | 0.015 | 0.013  | -0.013 | 0.025 |
| GPT-3.5 | 50%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-3.5 | 50%                   | Debiased         | 0.000  | 0.002 | 0.000  | -0.004 | 0.003 |
| GPT-4o  | 5%                    | LLM              | 0.005  | 0.008 | 0.007  | -0.008 | 0.016 |
| GPT-4o  | 5%                    | Human Validation | 0.001  | 0.002 | 0.001  | -0.003 | 0.005 |
| GPT-4o  | 5%                    | Debiased         | -0.001 | 0.004 | -0.001 | -0.007 | 0.005 |
| GPT-4o  | 10%                   | LLM              | 0.005  | 0.008 | 0.007  | -0.008 | 0.017 |
| GPT-4o  | 10%                   | Human Validation | 0.000  | 0.002 | 0.000  | -0.003 | 0.002 |
| GPT-4o  | 10%                   | Debiased         | 0.000  | 0.003 | 0.001  | -0.005 | 0.004 |
| GPT-4o  | 25%                   | LLM              | 0.004  | 0.008 | 0.006  | -0.008 | 0.015 |
| GPT-4o  | 25%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.002 |
| GPT-4o  | 25%                   | Debiased         | 0.000  | 0.002 | 0.000  | -0.002 | 0.003 |
| GPT-4o  | 50%                   | LLM              | 0.004  | 0.008 | 0.006  | -0.009 | 0.016 |
| GPT-4o  | 50%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-4o  | 50%                   | Debiased         | 0.000  | 0.002 | 0.000  | -0.003 | 0.002 |

Table 25: Bias Summary Statistics of  $\beta_{20}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | Bias             | Mean   | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|--------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.001  | 0.011 | 0.001  | -0.015 | 0.018 |
| GPT-3.5 | 5%                    | Human Validation | 0.000  | 0.003 | 0.000  | -0.006 | 0.005 |
| GPT-3.5 | 5%                    | Debiased         | 0.001  | 0.003 | 0.001  | -0.004 | 0.007 |
| GPT-3.5 | 10%                   | LLM              | 0.001  | 0.011 | 0.001  | -0.015 | 0.017 |
| GPT-3.5 | 10%                   | Human Validation | 0.000  | 0.002 | 0.000  | -0.002 | 0.003 |
| GPT-3.5 | 10%                   | Debiased         | 0.000  | 0.003 | 0.000  | -0.004 | 0.004 |
| GPT-3.5 | 25%                   | LLM              | 0.001  | 0.011 | 0.000  | -0.015 | 0.017 |
| GPT-3.5 | 25%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.002 |
| GPT-3.5 | 25%                   | Debiased         | 0.000  | 0.002 | 0.000  | -0.002 | 0.003 |
| GPT-3.5 | 50%                   | LLM              | 0.001  | 0.011 | 0.001  | -0.015 | 0.017 |
| GPT-3.5 | 50%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.002 |
| GPT-3.5 | 50%                   | Debiased         | 0.000  | 0.002 | 0.000  | -0.003 | 0.003 |
| GPT-4o  | 5%                    | LLM              | -0.003 | 0.014 | 0.001  | -0.022 | 0.012 |
| GPT-4o  | 5%                    | Human Validation | 0.000  | 0.003 | 0.000  | -0.004 | 0.004 |
| GPT-4o  | 5%                    | Debiased         | -0.001 | 0.002 | -0.001 | -0.004 | 0.003 |
| GPT-4o  | 10%                   | LLM              | -0.003 | 0.014 | 0.001  | -0.023 | 0.013 |
| GPT-4o  | 10%                   | Human Validation | 0.000  | 0.002 | 0.000  | -0.003 | 0.002 |
| GPT-4o  | 10%                   | Debiased         | 0.000  | 0.002 | 0.001  | -0.003 | 0.003 |
| GPT-4o  | 25%                   | LLM              | -0.003 | 0.014 | 0.001  | -0.022 | 0.012 |
| GPT-4o  | 25%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.002 |
| GPT-4o  | 25%                   | Debiased         | 0.000  | 0.001 | 0.000  | -0.002 | 0.001 |
| GPT-4o  | 50%                   | LLM              | -0.003 | 0.014 | 0.001  | -0.022 | 0.012 |
| GPT-4o  | 50%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.002 |
| GPT-4o  | 50%                   | Debiased         | 0.000  | 0.001 | 0.000  | -0.002 | 0.002 |

Table 26: Bias Summary Statistics of  $\beta_{\text{Other}}$  by Model and Proportion of Validation Samples for. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | Bias             | Mean   | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|--------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.000  | 0.010 | -0.002 | -0.013 | 0.014 |
| GPT-3.5 | 5%                    | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.002 |
| GPT-3.5 | 5%                    | Debiased         | -0.001 | 0.004 | -0.001 | -0.004 | 0.003 |
| GPT-3.5 | 10%                   | LLM              | 0.000  | 0.010 | -0.003 | -0.013 | 0.014 |
| GPT-3.5 | 10%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-3.5 | 10%                   | Debiased         | 0.000  | 0.001 | 0.000  | -0.002 | 0.002 |
| GPT-3.5 | 25%                   | LLM              | 0.000  | 0.010 | -0.002 | -0.013 | 0.014 |
| GPT-3.5 | 25%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-3.5 | 25%                   | Debiased         | 0.000  | 0.001 | 0.000  | -0.002 | 0.001 |
| GPT-3.5 | 50%                   | LLM              | 0.000  | 0.010 | -0.003 | -0.013 | 0.014 |
| GPT-3.5 | 50%                   | Human Validation | 0.000  | 0.000 | 0.000  | -0.001 | 0.001 |
| GPT-3.5 | 50%                   | Debiased         | 0.000  | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-4o  | 5%                    | LLM              | 0.000  | 0.005 | 0.001  | -0.007 | 0.007 |
| GPT-4o  | 5%                    | Human Validation | 0.000  | 0.001 | 0.000  | -0.002 | 0.002 |
| GPT-4o  | 5%                    | Debiased         | 0.000  | 0.002 | 0.000  | -0.002 | 0.003 |
| GPT-4o  | 10%                   | LLM              | 0.000  | 0.005 | 0.001  | -0.007 | 0.007 |
| GPT-4o  | 10%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-4o  | 10%                   | Debiased         | 0.000  | 0.001 | 0.000  | -0.002 | 0.002 |
| GPT-4o  | 25%                   | LLM              | 0.000  | 0.005 | 0.001  | -0.007 | 0.007 |
| GPT-4o  | 25%                   | Human Validation | 0.000  | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-4o  | 25%                   | Debiased         | 0.000  | 0.001 | 0.000  | -0.001 | 0.001 |
| GPT-4o  | 50%                   | LLM              | 0.000  | 0.005 | 0.001  | -0.007 | 0.007 |
| GPT-4o  | 50%                   | Human Validation | 0.000  | 0.000 | 0.000  | -0.001 | 0.001 |
| GPT-4o  | 50%                   | Debiased         | 0.000  | 0.001 | 0.000  | -0.001 | 0.001 |



Table 27: Normalized Bias Summary of  $\beta_3$  Statistics by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|--------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.212  | 1.089 | 0.483  | -1.480 | 1.463 |
| GPT-3.5 | 5%                    | Human Validation | 0.008  | 0.034 | 0.006  | -0.049 | 0.063 |
| GPT-3.5 | 5%                    | Debiased         | 0.001  | 0.028 | 0.001  | -0.042 | 0.048 |
| GPT-3.5 | 10%                   | LLM              | 0.225  | 1.085 | 0.500  | -1.453 | 1.500 |
| GPT-3.5 | 10%                   | Human Validation | 0.000  | 0.030 | -0.005 | -0.051 | 0.055 |
| GPT-3.5 | 10%                   | Debiased         | 0.005  | 0.028 | 0.006  | -0.032 | 0.046 |
| GPT-3.5 | 25%                   | LLM              | 0.230  | 1.094 | 0.538  | -1.462 | 1.493 |
| GPT-3.5 | 25%                   | Human Validation | 0.006  | 0.025 | 0.011  | -0.031 | 0.041 |
| GPT-3.5 | 25%                   | Debiased         | 0.002  | 0.034 | 0.001  | -0.045 | 0.057 |
| GPT-3.5 | 50%                   | LLM              | 0.222  | 1.091 | 0.515  | -1.490 | 1.519 |
| GPT-3.5 | 50%                   | Human Validation | -0.001 | 0.034 | -0.005 | -0.055 | 0.051 |
| GPT-3.5 | 50%                   | Debiased         | -0.006 | 0.027 | -0.004 | -0.050 | 0.038 |
| GPT-4o  | 5%                    | LLM              | 0.084  | 0.507 | -0.119 | -0.510 | 0.824 |
| GPT-4o  | 5%                    | Human Validation | -0.005 | 0.029 | -0.003 | -0.051 | 0.037 |
| GPT-4o  | 5%                    | Debiased         | 0.005  | 0.037 | 0.005  | -0.067 | 0.055 |
| GPT-4o  | 10%                   | LLM              | 0.075  | 0.501 | -0.135 | -0.468 | 0.825 |
| GPT-4o  | 10%                   | Human Validation | -0.005 | 0.028 | -0.007 | -0.043 | 0.042 |
| GPT-4o  | 10%                   | Debiased         | -0.006 | 0.036 | -0.006 | -0.074 | 0.052 |
| GPT-4o  | 25%                   | LLM              | 0.078  | 0.513 | -0.136 | -0.483 | 0.857 |
| GPT-4o  | 25%                   | Human Validation | 0.006  | 0.032 | 0.006  | -0.048 | 0.055 |
| GPT-4o  | 25%                   | Debiased         | -0.005 | 0.036 | -0.001 | -0.061 | 0.048 |
| GPT-4o  | 50%                   | LLM              | 0.078  | 0.497 | -0.119 | -0.514 | 0.805 |
| GPT-4o  | 50%                   | Human Validation | 0.003  | 0.029 | 0.013  | -0.041 | 0.038 |
| GPT-4o  | 50%                   | Debiased         | 0.001  | 0.032 | -0.004 | -0.041 | 0.066 |

Table 28: Normalized Bias Summary Statistics of  $\beta_{14}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|--------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | -0.003 | 1.821 | -0.848 | -2.053 | 2.819 |
| GPT-3.5 | 5%                    | Human Validation | -0.006 | 0.030 | -0.006 | -0.053 | 0.032 |
| GPT-3.5 | 5%                    | Debiased         | 0.003  | 0.035 | 0.000  | -0.057 | 0.061 |
| GPT-3.5 | 10%                   | LLM              | 0.003  | 1.826 | -0.853 | -2.059 | 2.826 |
| GPT-3.5 | 10%                   | Human Validation | -0.012 | 0.028 | -0.012 | -0.048 | 0.039 |
| GPT-3.5 | 10%                   | Debiased         | 0.002  | 0.043 | 0.009  | -0.062 | 0.061 |
| GPT-3.5 | 25%                   | LLM              | -0.021 | 1.822 | -0.808 | -2.035 | 2.855 |
| GPT-3.5 | 25%                   | Human Validation | 0.002  | 0.029 | 0.002  | -0.035 | 0.044 |
| GPT-3.5 | 25%                   | Debiased         | -0.001 | 0.034 | 0.006  | -0.068 | 0.039 |
| GPT-3.5 | 50%                   | LLM              | -0.003 | 1.832 | -0.886 | -2.059 | 2.815 |
| GPT-3.5 | 50%                   | Human Validation | 0.012  | 0.031 | 0.011  | -0.039 | 0.053 |
| GPT-3.5 | 50%                   | Debiased         | 0.005  | 0.031 | 0.001  | -0.042 | 0.053 |
| GPT-4o  | 5%                    | LLM              | 0.028  | 1.224 | -0.369 | -1.314 | 1.908 |
| GPT-4o  | 5%                    | Human Validation | 0.003  | 0.028 | 0.004  | -0.043 | 0.045 |
| GPT-4o  | 5%                    | Debiased         | 0.006  | 0.043 | 0.004  | -0.056 | 0.067 |
| GPT-4o  | 10%                   | LLM              | 0.023  | 1.224 | -0.370 | -1.344 | 1.890 |
| GPT-4o  | 10%                   | Human Validation | 0.000  | 0.027 | 0.001  | -0.040 | 0.044 |
| GPT-4o  | 10%                   | Debiased         | 0.001  | 0.035 | 0.002  | -0.046 | 0.073 |
| GPT-4o  | 25%                   | LLM              | 0.031  | 1.222 | -0.332 | -1.323 | 1.879 |
| GPT-4o  | 25%                   | Human Validation | 0.000  | 0.036 | -0.003 | -0.057 | 0.053 |
| GPT-4o  | 25%                   | Debiased         | 0.000  | 0.033 | 0.000  | -0.043 | 0.056 |
| GPT-4o  | 50%                   | LLM              | 0.022  | 1.223 | -0.414 | -1.309 | 1.859 |
| GPT-4o  | 50%                   | Human Validation | -0.006 | 0.030 | -0.010 | -0.043 | 0.047 |
| GPT-4o  | 50%                   | Debiased         | -0.004 | 0.038 | 0.000  | -0.061 | 0.055 |

Table 29: Normalized Bias Summary Statistics of  $\beta_{15}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|--------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.075  | 0.568 | 0.189  | -0.712 | 0.900 |
| GPT-3.5 | 5%                    | Human Validation | 0.000  | 0.033 | -0.007 | -0.052 | 0.056 |
| GPT-3.5 | 5%                    | Debiased         | 0.002  | 0.032 | 0.004  | -0.039 | 0.049 |
| GPT-3.5 | 10%                   | LLM              | 0.069  | 0.577 | 0.167  | -0.725 | 0.950 |
| GPT-3.5 | 10%                   | Human Validation | -0.002 | 0.035 | 0.002  | -0.041 | 0.047 |
| GPT-3.5 | 10%                   | Debiased         | -0.001 | 0.035 | -0.002 | -0.062 | 0.048 |
| GPT-3.5 | 25%                   | LLM              | 0.073  | 0.566 | 0.150  | -0.688 | 0.930 |
| GPT-3.5 | 25%                   | Human Validation | -0.011 | 0.031 | -0.015 | -0.056 | 0.043 |
| GPT-3.5 | 25%                   | Debiased         | 0.005  | 0.039 | 0.005  | -0.046 | 0.061 |
| GPT-3.5 | 50%                   | LLM              | 0.072  | 0.550 | 0.132  | -0.683 | 0.902 |
| GPT-3.5 | 50%                   | Human Validation | -0.007 | 0.025 | -0.002 | -0.048 | 0.032 |
| GPT-3.5 | 50%                   | Debiased         | -0.004 | 0.031 | -0.007 | -0.051 | 0.045 |
| GPT-4o  | 5%                    | LLM              | 0.160  | 0.328 | 0.140  | -0.276 | 0.693 |
| GPT-4o  | 5%                    | Human Validation | -0.006 | 0.033 | 0.000  | -0.055 | 0.048 |
| GPT-4o  | 5%                    | Debiased         | -0.006 | 0.028 | -0.006 | -0.050 | 0.033 |
| GPT-4o  | 10%                   | LLM              | 0.148  | 0.334 | 0.171  | -0.283 | 0.664 |
| GPT-4o  | 10%                   | Human Validation | 0.001  | 0.033 | -0.003 | -0.049 | 0.064 |
| GPT-4o  | 10%                   | Debiased         | -0.005 | 0.030 | -0.002 | -0.055 | 0.037 |
| GPT-4o  | 25%                   | LLM              | 0.169  | 0.322 | 0.152  | -0.241 | 0.685 |
| GPT-4o  | 25%                   | Human Validation | 0.001  | 0.037 | -0.006 | -0.045 | 0.071 |
| GPT-4o  | 25%                   | Debiased         | 0.004  | 0.031 | 0.010  | -0.056 | 0.044 |
| GPT-4o  | 50%                   | LLM              | 0.152  | 0.324 | 0.158  | -0.300 | 0.654 |
| GPT-4o  | 50%                   | Human Validation | -0.005 | 0.032 | -0.005 | -0.055 | 0.038 |
| GPT-4o  | 50%                   | Debiased         | 0.000  | 0.030 | 0.003  | -0.054 | 0.040 |

Table 30: Normalized Bias Summary Statistics of  $\beta_{19}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|--------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.448  | 0.723 | 0.594  | -0.586 | 1.374 |
| GPT-3.5 | 5%                    | Human Validation | 0.004  | 0.038 | 0.004  | -0.059 | 0.064 |
| GPT-3.5 | 5%                    | Debiased         | -0.009 | 0.035 | -0.008 | -0.062 | 0.033 |
| GPT-3.5 | 10%                   | LLM              | 0.452  | 0.728 | 0.665  | -0.575 | 1.387 |
| GPT-3.5 | 10%                   | Human Validation | -0.003 | 0.030 | -0.007 | -0.046 | 0.034 |
| GPT-3.5 | 10%                   | Debiased         | -0.007 | 0.034 | -0.008 | -0.061 | 0.048 |
| GPT-3.5 | 25%                   | LLM              | 0.443  | 0.723 | 0.642  | -0.572 | 1.402 |
| GPT-3.5 | 25%                   | Human Validation | -0.001 | 0.030 | -0.002 | -0.043 | 0.062 |
| GPT-3.5 | 25%                   | Debiased         | -0.002 | 0.034 | -0.004 | -0.051 | 0.046 |
| GPT-3.5 | 50%                   | LLM              | 0.447  | 0.737 | 0.622  | -0.620 | 1.413 |
| GPT-3.5 | 50%                   | Human Validation | 0.008  | 0.026 | 0.008  | -0.032 | 0.049 |
| GPT-3.5 | 50%                   | Debiased         | 0.002  | 0.036 | 0.005  | -0.058 | 0.052 |
| GPT-4o  | 5%                    | LLM              | 0.290  | 0.482 | 0.423  | -0.435 | 0.905 |
| GPT-4o  | 5%                    | Human Validation | 0.011  | 0.030 | 0.015  | -0.043 | 0.054 |
| GPT-4o  | 5%                    | Debiased         | -0.010 | 0.033 | -0.011 | -0.064 | 0.047 |
| GPT-4o  | 10%                   | LLM              | 0.285  | 0.471 | 0.412  | -0.420 | 0.914 |
| GPT-4o  | 10%                   | Human Validation | -0.004 | 0.028 | -0.006 | -0.043 | 0.041 |
| GPT-4o  | 10%                   | Debiased         | -0.001 | 0.035 | 0.010  | -0.064 | 0.046 |
| GPT-4o  | 25%                   | LLM              | 0.276  | 0.458 | 0.392  | -0.441 | 0.836 |
| GPT-4o  | 25%                   | Human Validation | -0.001 | 0.033 | 0.000  | -0.061 | 0.050 |
| GPT-4o  | 25%                   | Debiased         | 0.003  | 0.032 | 0.000  | -0.035 | 0.060 |
| GPT-4o  | 50%                   | LLM              | 0.275  | 0.468 | 0.382  | -0.447 | 0.859 |
| GPT-4o  | 50%                   | Human Validation | 0.005  | 0.029 | 0.005  | -0.034 | 0.041 |
| GPT-4o  | 50%                   | Debiased         | -0.006 | 0.033 | -0.005 | -0.068 | 0.041 |

Table 31: Normalized Bias Summary Statistics of  $\beta_{20}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|--------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.068  | 0.557 | 0.023  | -0.676 | 0.977 |
| GPT-3.5 | 5%                    | Human Validation | 0.001  | 0.038 | 0.003  | -0.070 | 0.052 |
| GPT-3.5 | 5%                    | Debiased         | 0.009  | 0.030 | 0.012  | -0.040 | 0.053 |
| GPT-3.5 | 10%                   | LLM              | 0.072  | 0.558 | 0.040  | -0.657 | 0.938 |
| GPT-3.5 | 10%                   | Human Validation | 0.005  | 0.032 | 0.001  | -0.038 | 0.055 |
| GPT-3.5 | 10%                   | Debiased         | 0.000  | 0.036 | 0.001  | -0.053 | 0.059 |
| GPT-3.5 | 25%                   | LLM              | 0.070  | 0.561 | 0.012  | -0.680 | 0.950 |
| GPT-3.5 | 25%                   | Human Validation | 0.012  | 0.028 | 0.011  | -0.031 | 0.057 |
| GPT-3.5 | 25%                   | Debiased         | 0.002  | 0.032 | -0.008 | -0.037 | 0.066 |
| GPT-3.5 | 50%                   | LLM              | 0.076  | 0.554 | 0.052  | -0.673 | 0.943 |
| GPT-3.5 | 50%                   | Human Validation | 0.004  | 0.029 | -0.002 | -0.037 | 0.053 |
| GPT-3.5 | 50%                   | Debiased         | -0.001 | 0.038 | 0.003  | -0.059 | 0.059 |
| GPT-4o  | 5%                    | LLM              | -0.085 | 0.697 | 0.038  | -1.051 | 0.757 |
| GPT-4o  | 5%                    | Human Validation | -0.003 | 0.031 | 0.001  | -0.050 | 0.042 |
| GPT-4o  | 5%                    | Debiased         | -0.006 | 0.028 | -0.010 | -0.039 | 0.041 |
| GPT-4o  | 10%                   | LLM              | -0.076 | 0.695 | 0.041  | -1.062 | 0.803 |
| GPT-4o  | 10%                   | Human Validation | -0.003 | 0.028 | 0.002  | -0.052 | 0.032 |
| GPT-4o  | 10%                   | Debiased         | 0.008  | 0.029 | 0.010  | -0.041 | 0.050 |
| GPT-4o  | 25%                   | LLM              | -0.087 | 0.677 | 0.042  | -1.030 | 0.752 |
| GPT-4o  | 25%                   | Human Validation | -0.005 | 0.031 | 0.001  | -0.057 | 0.044 |
| GPT-4o  | 25%                   | Debiased         | -0.003 | 0.029 | -0.007 | -0.044 | 0.041 |
| GPT-4o  | 50%                   | LLM              | -0.083 | 0.677 | 0.047  | -1.015 | 0.699 |
| GPT-4o  | 50%                   | Human Validation | -0.001 | 0.038 | 0.003  | -0.068 | 0.053 |
| GPT-4o  | 50%                   | Debiased         | 0.001  | 0.033 | 0.001  | -0.045 | 0.059 |

Table 32: Normalized Bias Summary Statistics of  $\beta_{\text{Other}}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | Normalized Bias  | Mean   | SD    | Median | 5%     | 95%   |
|---------|-----------------------|------------------|--------|-------|--------|--------|-------|
| GPT-3.5 | 5%                    | LLM              | -0.145 | 1.247 | -0.255 | -1.990 | 1.618 |
| GPT-3.5 | 5%                    | Human Validation | -0.009 | 0.028 | -0.010 | -0.055 | 0.040 |
| GPT-3.5 | 5%                    | Debiased         | -0.006 | 0.028 | -0.012 | -0.045 | 0.033 |
| GPT-3.5 | 10%                   | LLM              | -0.161 | 1.223 | -0.290 | -1.927 | 1.580 |
| GPT-3.5 | 10%                   | Human Validation | 0.001  | 0.031 | 0.004  | -0.048 | 0.045 |
| GPT-3.5 | 10%                   | Debiased         | 0.000  | 0.037 | -0.001 | -0.048 | 0.052 |
| GPT-3.5 | 25%                   | LLM              | -0.154 | 1.227 | -0.290 | -1.958 | 1.584 |
| GPT-3.5 | 25%                   | Human Validation | -0.004 | 0.030 | -0.003 | -0.059 | 0.043 |
| GPT-3.5 | 25%                   | Debiased         | -0.003 | 0.036 | -0.002 | -0.065 | 0.054 |
| GPT-3.5 | 50%                   | LLM              | -0.137 | 1.218 | -0.313 | -1.874 | 1.525 |
| GPT-3.5 | 50%                   | Human Validation | 0.000  | 0.034 | 0.006  | -0.056 | 0.043 |
| GPT-3.5 | 50%                   | Debiased         | 0.001  | 0.037 | -0.003 | -0.050 | 0.055 |
| GPT-4o  | 5%                    | LLM              | -0.041 | 0.620 | 0.099  | -1.010 | 0.731 |
| GPT-4o  | 5%                    | Human Validation | 0.002  | 0.036 | 0.005  | -0.047 | 0.060 |
| GPT-4o  | 5%                    | Debiased         | 0.009  | 0.034 | 0.007  | -0.039 | 0.057 |
| GPT-4o  | 10%                   | LLM              | -0.039 | 0.608 | 0.119  | -0.984 | 0.722 |
| GPT-4o  | 10%                   | Human Validation | -0.004 | 0.029 | 0.000  | -0.042 | 0.041 |
| GPT-4o  | 10%                   | Debiased         | 0.001  | 0.033 | -0.005 | -0.049 | 0.060 |
| GPT-4o  | 25%                   | LLM              | -0.047 | 0.630 | 0.100  | -1.064 | 0.736 |
| GPT-4o  | 25%                   | Human Validation | 0.007  | 0.030 | 0.003  | -0.033 | 0.055 |
| GPT-4o  | 25%                   | Debiased         | -0.008 | 0.037 | -0.007 | -0.060 | 0.052 |
| GPT-4o  | 50%                   | LLM              | -0.026 | 0.607 | 0.120  | -0.998 | 0.739 |
| GPT-4o  | 50%                   | Human Validation | -0.002 | 0.035 | 0.008  | -0.056 | 0.044 |
| GPT-4o  | 50%                   | Debiased         | 0.008  | 0.037 | 0.005  | -0.047 | 0.069 |

Table 33: MSE Summary Statistics of  $\beta_3$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|-----------------------|------------------|-------|-------|--------|-------|-------|
| 5%                    | LLM              | 0.001 | 0.001 | 0.001  | 0.001 | 0.002 |
| 5%                    | Human Validation | 0.012 | 0.002 | 0.013  | 0.009 | 0.014 |
| 5%                    | Debiased         | 0.015 | 0.015 | 0.011  | 0.006 | 0.031 |
| 10%                   | LLM              | 0.001 | 0.001 | 0.001  | 0.001 | 0.002 |
| 10%                   | Human Validation | 0.006 | 0.001 | 0.006  | 0.004 | 0.007 |
| 10%                   | Debiased         | 0.007 | 0.003 | 0.005  | 0.003 | 0.013 |
| 25%                   | LLM              | 0.001 | 0.001 | 0.001  | 0.001 | 0.003 |
| 25%                   | Human Validation | 0.002 | 0.000 | 0.002  | 0.002 | 0.003 |
| 25%                   | Debiased         | 0.003 | 0.001 | 0.003  | 0.002 | 0.006 |
| 50%                   | LLM              | 0.001 | 0.001 | 0.001  | 0.001 | 0.002 |
| 50%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
| 50%                   | Debiased         | 0.003 | 0.001 | 0.003  | 0.002 | 0.005 |

Table 34: MSE Summary Statistics of  $\beta_{14}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | MSE              | Mean  | SD     | Median | 5%    | 95%   |
|-----------------------|------------------|-------|--------|--------|-------|-------|
| 5%                    | LLM              | 0.001 | 0.001  | 0.001  | 0.001 | 0.003 |
|                       | Human Validation | 0.014 | 0.003  | 0.015  | 0.010 | 0.017 |
|                       | Debiased         | 3.543 | 23.671 | 0.099  | 0.055 | 2.287 |
| 10%                   | LLM              | 0.001 | 0.001  | 0.001  | 0.001 | 0.003 |
|                       | Human Validation | 0.007 | 0.001  | 0.007  | 0.005 | 0.009 |
|                       | Debiased         | 0.037 | 0.013  | 0.036  | 0.020 | 0.057 |
| 25%                   | LLM              | 0.001 | 0.001  | 0.001  | 0.001 | 0.003 |
|                       | Human Validation | 0.003 | 0.001  | 0.003  | 0.002 | 0.003 |
|                       | Debiased         | 0.016 | 0.005  | 0.015  | 0.009 | 0.026 |
| 50%                   | LLM              | 0.001 | 0.001  | 0.001  | 0.001 | 0.003 |
|                       | Human Validation | 0.001 | 0.000  | 0.001  | 0.001 | 0.002 |
|                       | Debiased         | 0.012 | 0.004  | 0.012  | 0.007 | 0.018 |

Table 35: MSE Summary Statistics of  $\beta_{15}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|-----------------------|------------------|-------|-------|--------|-------|-------|
| 5%                    | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Human Validation | 0.008 | 0.002 | 0.009  | 0.005 | 0.011 |
|                       | Debiased         | 0.018 | 0.006 | 0.018  | 0.009 | 0.029 |
| 10%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Human Validation | 0.004 | 0.001 | 0.004  | 0.002 | 0.005 |
|                       | Debiased         | 0.008 | 0.003 | 0.009  | 0.004 | 0.013 |
| 25%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Human Validation | 0.002 | 0.000 | 0.002  | 0.001 | 0.002 |
|                       | Debiased         | 0.004 | 0.001 | 0.004  | 0.002 | 0.006 |
| 50%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Human Validation | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Debiased         | 0.003 | 0.001 | 0.004  | 0.002 | 0.005 |

Table 36: MSE Summary Statistics of  $\beta_{19}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|-----------------------|------------------|-------|-------|--------|-------|-------|
| 5%                    | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Human Validation | 0.006 | 0.001 | 0.006  | 0.004 | 0.008 |
|                       | Debiased         | 0.061 | 0.369 | 0.015  | 0.008 | 0.038 |
| 10%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Human Validation | 0.003 | 0.001 | 0.003  | 0.002 | 0.004 |
|                       | Debiased         | 0.008 | 0.003 | 0.007  | 0.004 | 0.012 |
| 25%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Human Validation | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
|                       | Debiased         | 0.004 | 0.001 | 0.003  | 0.002 | 0.006 |
| 50%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Human Validation | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Debiased         | 0.003 | 0.001 | 0.003  | 0.002 | 0.005 |

Table 37: MSE Summary Statistics of  $\beta_{20}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|-----------------------|------------------|-------|-------|--------|-------|-------|
| 5%                    | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Human Validation | 0.007 | 0.002 | 0.008  | 0.005 | 0.009 |
|                       | Debiased         | 0.009 | 0.006 | 0.009  | 0.004 | 0.015 |
| 10%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Human Validation | 0.004 | 0.001 | 0.004  | 0.002 | 0.005 |
|                       | Debiased         | 0.004 | 0.001 | 0.004  | 0.002 | 0.007 |
| 25%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Human Validation | 0.001 | 0.000 | 0.002  | 0.001 | 0.002 |
|                       | Debiased         | 0.002 | 0.001 | 0.002  | 0.001 | 0.004 |
| 50%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Human Validation | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
|                       | Debiased         | 0.002 | 0.001 | 0.002  | 0.001 | 0.003 |

Table 38: MSE Summary Statistics of  $\beta_{\text{Other}}$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|-----------------------|------------------|-------|-------|--------|-------|-------|
| 5%                    | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
|                       | Human Validation | 0.002 | 0.000 | 0.002  | 0.001 | 0.002 |
|                       | Debiased         | 0.011 | 0.052 | 0.002  | 0.001 | 0.009 |
| 10%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
|                       | Human Validation | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
|                       | Debiased         | 0.001 | 0.000 | 0.001  | 0.001 | 0.002 |
| 25%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
|                       | Human Validation | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
|                       | Debiased         | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| 50%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
|                       | Human Validation | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
|                       | Debiased         | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |



Table 39: MSE Summary Statistics of  $\beta_3$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|-------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.001 | 0.001 | 0.001  | 0.001 | 0.003 |
| GPT-3.5 | 5%                    | Human Validation | 0.012 | 0.002 | 0.013  | 0.009 | 0.014 |
| GPT-3.5 | 5%                    | Debiased         | 0.022 | 0.019 | 0.019  | 0.008 | 0.032 |
| GPT-3.5 | 10%                   | LLM              | 0.001 | 0.001 | 0.002  | 0.001 | 0.003 |
| GPT-3.5 | 10%                   | Human Validation | 0.006 | 0.001 | 0.006  | 0.004 | 0.007 |
| GPT-3.5 | 10%                   | Debiased         | 0.009 | 0.003 | 0.008  | 0.004 | 0.014 |
| GPT-3.5 | 25%                   | LLM              | 0.001 | 0.001 | 0.001  | 0.001 | 0.003 |
| GPT-3.5 | 25%                   | Human Validation | 0.002 | 0.000 | 0.002  | 0.002 | 0.003 |
| GPT-3.5 | 25%                   | Debiased         | 0.004 | 0.001 | 0.004  | 0.002 | 0.007 |
| GPT-3.5 | 50%                   | LLM              | 0.001 | 0.001 | 0.002  | 0.001 | 0.003 |
| GPT-3.5 | 50%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
| GPT-3.5 | 50%                   | Debiased         | 0.004 | 0.001 | 0.004  | 0.002 | 0.006 |
| GPT-4o  | 5%                    | LLM              | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
| GPT-4o  | 5%                    | Human Validation | 0.012 | 0.002 | 0.013  | 0.009 | 0.014 |
| GPT-4o  | 5%                    | Debiased         | 0.009 | 0.002 | 0.009  | 0.006 | 0.011 |
| GPT-4o  | 10%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
| GPT-4o  | 10%                   | Human Validation | 0.006 | 0.001 | 0.006  | 0.004 | 0.007 |
| GPT-4o  | 10%                   | Debiased         | 0.004 | 0.001 | 0.004  | 0.003 | 0.005 |
| GPT-4o  | 25%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
| GPT-4o  | 25%                   | Human Validation | 0.002 | 0.000 | 0.002  | 0.002 | 0.003 |
| GPT-4o  | 25%                   | Debiased         | 0.002 | 0.000 | 0.003  | 0.002 | 0.003 |
| GPT-4o  | 50%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
| GPT-4o  | 50%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
| GPT-4o  | 50%                   | Debiased         | 0.002 | 0.000 | 0.002  | 0.002 | 0.003 |

Table 40: MSE Summary Statistics of  $\beta_{14}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | MSE              | Mean  | SD     | Median | 5%    | 95%    |
|---------|-----------------------|------------------|-------|--------|--------|-------|--------|
| GPT-3.5 | 5%                    | LLM              | 0.002 | 0.001  | 0.002  | 0.001 | 0.003  |
| GPT-3.5 | 5%                    | Human Validation | 0.014 | 0.003  | 0.015  | 0.010 | 0.018  |
| GPT-3.5 | 5%                    | Debiased         | 6.812 | 33.377 | 0.160  | 0.083 | 11.343 |
| GPT-3.5 | 10%                   | LLM              | 0.002 | 0.001  | 0.002  | 0.001 | 0.003  |
| GPT-3.5 | 10%                   | Human Validation | 0.007 | 0.001  | 0.007  | 0.005 | 0.008  |
| GPT-3.5 | 10%                   | Debiased         | 0.046 | 0.011  | 0.048  | 0.030 | 0.060  |
| GPT-3.5 | 25%                   | LLM              | 0.002 | 0.001  | 0.002  | 0.001 | 0.003  |
| GPT-3.5 | 25%                   | Human Validation | 0.003 | 0.001  | 0.003  | 0.002 | 0.003  |
| GPT-3.5 | 25%                   | Debiased         | 0.020 | 0.005  | 0.020  | 0.013 | 0.026  |
| GPT-3.5 | 50%                   | LLM              | 0.002 | 0.001  | 0.002  | 0.001 | 0.003  |
| GPT-3.5 | 50%                   | Human Validation | 0.001 | 0.000  | 0.001  | 0.001 | 0.002  |
| GPT-3.5 | 50%                   | Debiased         | 0.014 | 0.003  | 0.015  | 0.009 | 0.019  |
| GPT-4o  | 5%                    | LLM              | 0.001 | 0.000  | 0.001  | 0.001 | 0.002  |
| GPT-4o  | 5%                    | Human Validation | 0.014 | 0.003  | 0.015  | 0.010 | 0.017  |
| GPT-4o  | 5%                    | Debiased         | 0.273 | 0.804  | 0.080  | 0.050 | 0.778  |
| GPT-4o  | 10%                   | LLM              | 0.001 | 0.000  | 0.001  | 0.001 | 0.002  |
| GPT-4o  | 10%                   | Human Validation | 0.007 | 0.001  | 0.007  | 0.005 | 0.009  |
| GPT-4o  | 10%                   | Debiased         | 0.029 | 0.008  | 0.031  | 0.019 | 0.039  |
| GPT-4o  | 25%                   | LLM              | 0.001 | 0.000  | 0.001  | 0.001 | 0.002  |
| GPT-4o  | 25%                   | Human Validation | 0.003 | 0.001  | 0.003  | 0.002 | 0.003  |
| GPT-4o  | 25%                   | Debiased         | 0.013 | 0.003  | 0.013  | 0.009 | 0.017  |
| GPT-4o  | 50%                   | LLM              | 0.001 | 0.000  | 0.001  | 0.001 | 0.002  |
| GPT-4o  | 50%                   | Human Validation | 0.001 | 0.000  | 0.001  | 0.001 | 0.002  |
| GPT-4o  | 50%                   | Debiased         | 0.010 | 0.002  | 0.010  | 0.007 | 0.013  |

Table 41: MSE Summary Statistics of  $\beta_{15}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|-------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 5%                    | Human Validation | 0.008 | 0.002 | 0.009  | 0.005 | 0.011 |
| GPT-3.5 | 5%                    | Debiased         | 0.020 | 0.007 | 0.019  | 0.010 | 0.030 |
| GPT-3.5 | 10%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 10%                   | Human Validation | 0.004 | 0.001 | 0.004  | 0.002 | 0.005 |
| GPT-3.5 | 10%                   | Debiased         | 0.009 | 0.003 | 0.009  | 0.004 | 0.013 |
| GPT-3.5 | 25%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 25%                   | Human Validation | 0.002 | 0.000 | 0.002  | 0.001 | 0.002 |
| GPT-3.5 | 25%                   | Debiased         | 0.004 | 0.001 | 0.004  | 0.002 | 0.007 |
| GPT-3.5 | 50%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 50%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 50%                   | Debiased         | 0.004 | 0.001 | 0.004  | 0.002 | 0.005 |
| GPT-4o  | 5%                    | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 5%                    | Human Validation | 0.008 | 0.002 | 0.009  | 0.005 | 0.011 |
| GPT-4o  | 5%                    | Debiased         | 0.016 | 0.005 | 0.018  | 0.009 | 0.023 |
| GPT-4o  | 10%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 10%                   | Human Validation | 0.004 | 0.001 | 0.004  | 0.002 | 0.005 |
| GPT-4o  | 10%                   | Debiased         | 0.008 | 0.002 | 0.008  | 0.004 | 0.011 |
| GPT-4o  | 25%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 25%                   | Human Validation | 0.002 | 0.000 | 0.002  | 0.001 | 0.002 |
| GPT-4o  | 25%                   | Debiased         | 0.004 | 0.001 | 0.004  | 0.002 | 0.005 |
| GPT-4o  | 50%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 50%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 50%                   | Debiased         | 0.003 | 0.001 | 0.004  | 0.002 | 0.005 |

Table 42: MSE Summary Statistics of  $\beta_{19}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|-------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 5%                    | Human Validation | 0.006 | 0.001 | 0.006  | 0.004 | 0.008 |
| GPT-3.5 | 5%                    | Debiased         | 0.108 | 0.521 | 0.021  | 0.012 | 0.042 |
| GPT-3.5 | 10%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 10%                   | Human Validation | 0.003 | 0.001 | 0.003  | 0.002 | 0.004 |
| GPT-3.5 | 10%                   | Debiased         | 0.009 | 0.003 | 0.009  | 0.006 | 0.013 |
| GPT-3.5 | 25%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 25%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
| GPT-3.5 | 25%                   | Debiased         | 0.004 | 0.001 | 0.005  | 0.003 | 0.006 |
| GPT-3.5 | 50%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 50%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 50%                   | Debiased         | 0.004 | 0.001 | 0.004  | 0.002 | 0.005 |
| GPT-4o  | 5%                    | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| GPT-4o  | 5%                    | Human Validation | 0.006 | 0.001 | 0.006  | 0.004 | 0.008 |
| GPT-4o  | 5%                    | Debiased         | 0.013 | 0.005 | 0.013  | 0.008 | 0.018 |
| GPT-4o  | 10%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| GPT-4o  | 10%                   | Human Validation | 0.003 | 0.001 | 0.003  | 0.002 | 0.004 |
| GPT-4o  | 10%                   | Debiased         | 0.006 | 0.001 | 0.006  | 0.004 | 0.007 |
| GPT-4o  | 25%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| GPT-4o  | 25%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
| GPT-4o  | 25%                   | Debiased         | 0.003 | 0.001 | 0.003  | 0.002 | 0.004 |
| GPT-4o  | 50%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |
| GPT-4o  | 50%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 50%                   | Debiased         | 0.002 | 0.001 | 0.003  | 0.002 | 0.003 |

Table 43: MSE Summary Statistics of  $\beta_{20}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|-------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 5%                    | Human Validation | 0.007 | 0.002 | 0.008  | 0.005 | 0.009 |
| GPT-3.5 | 5%                    | Debiased         | 0.011 | 0.008 | 0.010  | 0.005 | 0.016 |
| GPT-3.5 | 10%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 10%                   | Human Validation | 0.004 | 0.001 | 0.004  | 0.002 | 0.005 |
| GPT-3.5 | 10%                   | Debiased         | 0.005 | 0.002 | 0.005  | 0.003 | 0.008 |
| GPT-3.5 | 25%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 25%                   | Human Validation | 0.001 | 0.000 | 0.002  | 0.001 | 0.002 |
| GPT-3.5 | 25%                   | Debiased         | 0.003 | 0.001 | 0.003  | 0.001 | 0.004 |
| GPT-3.5 | 50%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 50%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 50%                   | Debiased         | 0.002 | 0.001 | 0.002  | 0.001 | 0.003 |
| GPT-4o  | 5%                    | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 5%                    | Human Validation | 0.007 | 0.002 | 0.008  | 0.005 | 0.009 |
| GPT-4o  | 5%                    | Debiased         | 0.007 | 0.002 | 0.008  | 0.004 | 0.010 |
| GPT-4o  | 10%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 10%                   | Human Validation | 0.004 | 0.001 | 0.004  | 0.002 | 0.005 |
| GPT-4o  | 10%                   | Debiased         | 0.004 | 0.001 | 0.004  | 0.002 | 0.005 |
| GPT-4o  | 25%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 25%                   | Human Validation | 0.001 | 0.000 | 0.002  | 0.001 | 0.002 |
| GPT-4o  | 25%                   | Debiased         | 0.002 | 0.001 | 0.002  | 0.001 | 0.003 |
| GPT-4o  | 50%                   | LLM              | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 50%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 50%                   | Debiased         | 0.002 | 0.001 | 0.002  | 0.001 | 0.002 |

Table 44: MSE Summary Statistics of  $\beta_{\text{Other}}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|-------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-3.5 | 5%                    | Human Validation | 0.002 | 0.000 | 0.002  | 0.001 | 0.002 |
| GPT-3.5 | 5%                    | Debiased         | 0.019 | 0.073 | 0.003  | 0.002 | 0.047 |
| GPT-3.5 | 10%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-3.5 | 10%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
| GPT-3.5 | 10%                   | Debiased         | 0.001 | 0.000 | 0.001  | 0.001 | 0.002 |
| GPT-3.5 | 25%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-3.5 | 25%                   | Human Validation | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-3.5 | 25%                   | Debiased         | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-3.5 | 50%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-3.5 | 50%                   | Human Validation | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-3.5 | 50%                   | Debiased         | 0.001 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 5%                    | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-4o  | 5%                    | Human Validation | 0.002 | 0.000 | 0.002  | 0.001 | 0.002 |
| GPT-4o  | 5%                    | Debiased         | 0.002 | 0.002 | 0.002  | 0.001 | 0.005 |
| GPT-4o  | 10%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-4o  | 10%                   | Human Validation | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
| GPT-4o  | 10%                   | Debiased         | 0.001 | 0.000 | 0.001  | 0.001 | 0.001 |
| GPT-4o  | 25%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-4o  | 25%                   | Human Validation | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-4o  | 25%                   | Debiased         | 0.000 | 0.000 | 0.001  | 0.000 | 0.001 |
| GPT-4o  | 50%                   | LLM              | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-4o  | 50%                   | Human Validation | 0.000 | 0.000 | 0.000  | 0.000 | 0.000 |
| GPT-4o  | 50%                   | Debiased         | 0.000 | 0.000 | 0.000  | 0.000 | 0.001 |

Table 45: Coverage Summary Statistics of  $\beta_3$  by Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Coverage | Mean             | SD    | Median | 5%    | 95%   |       |
|-----------------------|----------|------------------|-------|--------|-------|-------|-------|
|                       | 5%       | LLM              | 0.867 | 0.096  | 0.911 | 0.689 | 0.952 |
|                       | 5%       | Human Validation | 0.919 | 0.008  | 0.919 | 0.906 | 0.931 |
|                       | 5%       | Debiased         | 0.920 | 0.011  | 0.921 | 0.904 | 0.939 |
|                       | 10%      | LLM              | 0.866 | 0.097  | 0.913 | 0.679 | 0.947 |
|                       | 10%      | Human Validation | 0.935 | 0.008  | 0.935 | 0.923 | 0.947 |
|                       | 10%      | Debiased         | 0.940 | 0.007  | 0.940 | 0.928 | 0.952 |
|                       | 25%      | LLM              | 0.865 | 0.097  | 0.909 | 0.692 | 0.948 |
|                       | 25%      | Human Validation | 0.944 | 0.008  | 0.944 | 0.931 | 0.954 |
|                       | 25%      | Debiased         | 0.945 | 0.007  | 0.946 | 0.934 | 0.955 |
|                       | 50%      | LLM              | 0.865 | 0.099  | 0.913 | 0.653 | 0.947 |
|                       | 50%      | Human Validation | 0.949 | 0.007  | 0.948 | 0.939 | 0.959 |
|                       | 50%      | Debiased         | 0.948 | 0.007  | 0.949 | 0.937 | 0.957 |

Table 46: Coverage Summary Statistics of  $\beta_{14}$  by Proportion of Validation Samples. Proxy on the RHS:  
 $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Coverage         | Mean  | SD    | Median | 5%    | 95%   |
|-----------------------|------------------|-------|-------|--------|-------|-------|
| 5%                    | LLM              | 0.685 | 0.225 | 0.764  | 0.252 | 0.932 |
| 5%                    | Human Validation | 0.917 | 0.009 | 0.917  | 0.901 | 0.930 |
| 5%                    | Debiased         | 0.938 | 0.010 | 0.939  | 0.920 | 0.953 |
| 10%                   | LLM              | 0.689 | 0.226 | 0.770  | 0.242 | 0.937 |
| 10%                   | Human Validation | 0.933 | 0.008 | 0.934  | 0.918 | 0.945 |
| 10%                   | Debiased         | 0.940 | 0.008 | 0.942  | 0.926 | 0.951 |
| 25%                   | LLM              | 0.686 | 0.225 | 0.759  | 0.237 | 0.938 |
| 25%                   | Human Validation | 0.943 | 0.008 | 0.944  | 0.929 | 0.955 |
| 25%                   | Debiased         | 0.946 | 0.007 | 0.946  | 0.937 | 0.957 |
| 50%                   | LLM              | 0.690 | 0.225 | 0.766  | 0.255 | 0.935 |
| 50%                   | Human Validation | 0.947 | 0.007 | 0.948  | 0.936 | 0.959 |
| 50%                   | Debiased         | 0.948 | 0.007 | 0.949  | 0.937 | 0.958 |

Table 47: Coverage Summary Statistics of  $\beta_{15}$  by Proportion of Validation Samples. Proxy on the RHS:  
 $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Coverage         | Mean  | SD    | Median | 5%    | 95%   |
|-----------------------|------------------|-------|-------|--------|-------|-------|
| 5%                    | LLM              | 0.925 | 0.032 | 0.936  | 0.879 | 0.957 |
| 5%                    | Human Validation | 0.932 | 0.009 | 0.931  | 0.919 | 0.946 |
| 5%                    | Debiased         | 0.932 | 0.010 | 0.933  | 0.919 | 0.950 |
| 10%                   | LLM              | 0.926 | 0.030 | 0.938  | 0.876 | 0.953 |
| 10%                   | Human Validation | 0.940 | 0.008 | 0.941  | 0.927 | 0.951 |
| 10%                   | Debiased         | 0.941 | 0.008 | 0.941  | 0.928 | 0.955 |
| 25%                   | LLM              | 0.925 | 0.033 | 0.938  | 0.868 | 0.953 |
| 25%                   | Human Validation | 0.946 | 0.007 | 0.946  | 0.934 | 0.958 |
| 25%                   | Debiased         | 0.947 | 0.007 | 0.947  | 0.934 | 0.957 |
| 50%                   | LLM              | 0.926 | 0.031 | 0.933  | 0.883 | 0.956 |
| 50%                   | Human Validation | 0.947 | 0.006 | 0.947  | 0.938 | 0.958 |
| 50%                   | Debiased         | 0.947 | 0.007 | 0.947  | 0.934 | 0.957 |

Table 48: Coverage Summary Statistics of  $\beta_{19}$  by Proportion of Validation Samples. Proxy on the RHS:  
 $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Coverage         | Mean  | SD    | Median | 5%    | 95%   |
|-----------------------|------------------|-------|-------|--------|-------|-------|
| 5%                    | LLM              | 0.891 | 0.065 | 0.915  | 0.740 | 0.954 |
| 5%                    | Human Validation | 0.937 | 0.009 | 0.938  | 0.926 | 0.953 |
| 5%                    | Debiased         | 0.944 | 0.009 | 0.944  | 0.931 | 0.956 |
| 10%                   | LLM              | 0.890 | 0.066 | 0.912  | 0.750 | 0.949 |
| 10%                   | Human Validation | 0.943 | 0.008 | 0.943  | 0.930 | 0.954 |
| 10%                   | Debiased         | 0.947 | 0.008 | 0.947  | 0.934 | 0.957 |
| 25%                   | LLM              | 0.894 | 0.064 | 0.915  | 0.737 | 0.954 |
| 25%                   | Human Validation | 0.947 | 0.007 | 0.947  | 0.938 | 0.958 |
| 25%                   | Debiased         | 0.947 | 0.008 | 0.948  | 0.935 | 0.959 |
| 50%                   | LLM              | 0.889 | 0.066 | 0.913  | 0.735 | 0.947 |
| 50%                   | Human Validation | 0.951 | 0.007 | 0.950  | 0.940 | 0.961 |
| 50%                   | Debiased         | 0.947 | 0.007 | 0.948  | 0.935 | 0.957 |

Table 49: Coverage Summary Statistics of  $\beta_{20}$  by Proportion of Validation Samples. Proxy on the RHS:  
 $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Coverage             | Mean  | SD    | Median | 5%    | 95%   |
|-----------------------|----------------------|-------|-------|--------|-------|-------|
|                       | 5% LLM               | 0.901 | 0.062 | 0.923  | 0.784 | 0.956 |
|                       | 5% Human Validation  | 0.938 | 0.007 | 0.938  | 0.926 | 0.950 |
|                       | 5% Debiased          | 0.938 | 0.008 | 0.938  | 0.926 | 0.951 |
|                       | 10% LLM              | 0.902 | 0.056 | 0.920  | 0.785 | 0.953 |
|                       | 10% Human Validation | 0.943 | 0.007 | 0.942  | 0.932 | 0.956 |
|                       | 10% Debiased         | 0.945 | 0.007 | 0.944  | 0.934 | 0.956 |
|                       | 25% LLM              | 0.903 | 0.058 | 0.925  | 0.783 | 0.952 |
|                       | 25% Human Validation | 0.948 | 0.007 | 0.948  | 0.937 | 0.959 |
|                       | 25% Debiased         | 0.947 | 0.007 | 0.947  | 0.937 | 0.959 |
|                       | 50% LLM              | 0.904 | 0.055 | 0.923  | 0.796 | 0.954 |
|                       | 50% Human Validation | 0.950 | 0.007 | 0.951  | 0.938 | 0.960 |
|                       | 50% Debiased         | 0.949 | 0.007 | 0.948  | 0.939 | 0.961 |

Table 50: Coverage Summary Statistics of  $\beta_{\text{Other}}$  by Proportion of Validation Samples. Proxy on the RHS:  
 $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Validation Proportion | Coverage             | Mean  | SD    | Median | 5%    | 95%   |
|-----------------------|----------------------|-------|-------|--------|-------|-------|
|                       | 5% LLM               | 0.836 | 0.133 | 0.886  | 0.570 | 0.950 |
|                       | 5% Human Validation  | 0.949 | 0.007 | 0.949  | 0.937 | 0.961 |
|                       | 5% Debiased          | 0.954 | 0.007 | 0.954  | 0.943 | 0.966 |
|                       | 10% LLM              | 0.841 | 0.133 | 0.891  | 0.550 | 0.955 |
|                       | 10% Human Validation | 0.950 | 0.007 | 0.950  | 0.939 | 0.960 |
|                       | 10% Debiased         | 0.951 | 0.008 | 0.952  | 0.938 | 0.962 |
|                       | 25% LLM              | 0.839 | 0.133 | 0.887  | 0.581 | 0.949 |
|                       | 25% Human Validation | 0.949 | 0.007 | 0.950  | 0.938 | 0.958 |
|                       | 25% Debiased         | 0.948 | 0.007 | 0.950  | 0.935 | 0.959 |
|                       | 50% LLM              | 0.840 | 0.129 | 0.891  | 0.561 | 0.951 |
|                       | 50% Human Validation | 0.951 | 0.007 | 0.950  | 0.940 | 0.962 |
|                       | 50% Debiased         | 0.949 | 0.007 | 0.949  | 0.939 | 0.960 |



Table 51: Coverage Summary Statistics of  $\beta_3$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|-------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.812 | 0.108 | 0.804  | 0.642 | 0.938 |
| GPT-3.5 | 5%                    | Human Validation | 0.918 | 0.008 | 0.917  | 0.905 | 0.931 |
| GPT-3.5 | 5%                    | Debiased         | 0.926 | 0.010 | 0.927  | 0.913 | 0.941 |
| GPT-3.5 | 10%                   | LLM              | 0.811 | 0.110 | 0.810  | 0.636 | 0.934 |
| GPT-3.5 | 10%                   | Human Validation | 0.935 | 0.007 | 0.937  | 0.923 | 0.945 |
| GPT-3.5 | 10%                   | Debiased         | 0.940 | 0.009 | 0.940  | 0.925 | 0.955 |
| GPT-3.5 | 25%                   | LLM              | 0.810 | 0.109 | 0.800  | 0.623 | 0.936 |
| GPT-3.5 | 25%                   | Human Validation | 0.942 | 0.007 | 0.942  | 0.929 | 0.955 |
| GPT-3.5 | 25%                   | Debiased         | 0.946 | 0.006 | 0.946  | 0.934 | 0.955 |
| GPT-3.5 | 50%                   | LLM              | 0.809 | 0.113 | 0.820  | 0.621 | 0.937 |
| GPT-3.5 | 50%                   | Human Validation | 0.949 | 0.008 | 0.948  | 0.938 | 0.961 |
| GPT-3.5 | 50%                   | Debiased         | 0.946 | 0.007 | 0.946  | 0.937 | 0.956 |
| GPT-4o  | 5%                    | LLM              | 0.923 | 0.030 | 0.932  | 0.874 | 0.958 |
| GPT-4o  | 5%                    | Human Validation | 0.920 | 0.008 | 0.920  | 0.908 | 0.933 |
| GPT-4o  | 5%                    | Debiased         | 0.914 | 0.009 | 0.913  | 0.901 | 0.928 |
| GPT-4o  | 10%                   | LLM              | 0.922 | 0.027 | 0.933  | 0.871 | 0.950 |
| GPT-4o  | 10%                   | Human Validation | 0.935 | 0.008 | 0.935  | 0.923 | 0.948 |
| GPT-4o  | 10%                   | Debiased         | 0.940 | 0.005 | 0.940  | 0.929 | 0.947 |
| GPT-4o  | 25%                   | LLM              | 0.921 | 0.030 | 0.934  | 0.875 | 0.951 |
| GPT-4o  | 25%                   | Human Validation | 0.946 | 0.007 | 0.946  | 0.933 | 0.954 |
| GPT-4o  | 25%                   | Debiased         | 0.945 | 0.007 | 0.946  | 0.933 | 0.954 |
| GPT-4o  | 50%                   | LLM              | 0.921 | 0.027 | 0.929  | 0.874 | 0.948 |
| GPT-4o  | 50%                   | Human Validation | 0.948 | 0.007 | 0.948  | 0.939 | 0.958 |
| GPT-4o  | 50%                   | Debiased         | 0.949 | 0.006 | 0.951  | 0.940 | 0.957 |

Table 52: Coverage Summary Statistics of  $\beta_{14}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|-------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.596 | 0.255 | 0.650  | 0.178 | 0.903 |
| GPT-3.5 | 5%                    | Human Validation | 0.917 | 0.010 | 0.917  | 0.901 | 0.933 |
| GPT-3.5 | 5%                    | Debiased         | 0.944 | 0.006 | 0.944  | 0.936 | 0.956 |
| GPT-3.5 | 10%                   | LLM              | 0.599 | 0.255 | 0.667  | 0.200 | 0.913 |
| GPT-3.5 | 10%                   | Human Validation | 0.932 | 0.008 | 0.932  | 0.916 | 0.944 |
| GPT-3.5 | 10%                   | Debiased         | 0.942 | 0.010 | 0.943  | 0.926 | 0.953 |
| GPT-3.5 | 25%                   | LLM              | 0.599 | 0.256 | 0.659  | 0.172 | 0.915 |
| GPT-3.5 | 25%                   | Human Validation | 0.943 | 0.008 | 0.944  | 0.931 | 0.953 |
| GPT-3.5 | 25%                   | Debiased         | 0.946 | 0.009 | 0.947  | 0.933 | 0.959 |
| GPT-3.5 | 50%                   | LLM              | 0.602 | 0.255 | 0.649  | 0.184 | 0.921 |
| GPT-3.5 | 50%                   | Human Validation | 0.947 | 0.007 | 0.948  | 0.937 | 0.959 |
| GPT-3.5 | 50%                   | Debiased         | 0.948 | 0.006 | 0.949  | 0.939 | 0.957 |
| GPT-4o  | 5%                    | LLM              | 0.775 | 0.147 | 0.777  | 0.540 | 0.941 |
| GPT-4o  | 5%                    | Human Validation | 0.917 | 0.008 | 0.917  | 0.905 | 0.928 |
| GPT-4o  | 5%                    | Debiased         | 0.933 | 0.010 | 0.935  | 0.920 | 0.946 |
| GPT-4o  | 10%                   | LLM              | 0.779 | 0.150 | 0.792  | 0.524 | 0.950 |
| GPT-4o  | 10%                   | Human Validation | 0.934 | 0.008 | 0.935  | 0.919 | 0.945 |
| GPT-4o  | 10%                   | Debiased         | 0.939 | 0.007 | 0.940  | 0.928 | 0.949 |
| GPT-4o  | 25%                   | LLM              | 0.774 | 0.146 | 0.786  | 0.536 | 0.940 |
| GPT-4o  | 25%                   | Human Validation | 0.943 | 0.008 | 0.944  | 0.928 | 0.955 |
| GPT-4o  | 25%                   | Debiased         | 0.946 | 0.006 | 0.944  | 0.938 | 0.957 |
| GPT-4o  | 50%                   | LLM              | 0.777 | 0.148 | 0.793  | 0.537 | 0.939 |
| GPT-4o  | 50%                   | Human Validation | 0.947 | 0.008 | 0.948  | 0.934 | 0.958 |
| GPT-4o  | 50%                   | Debiased         | 0.948 | 0.008 | 0.949  | 0.936 | 0.959 |

Table 53: Coverage Summary Statistics of  $\beta_{15}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|-------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.914 | 0.038 | 0.923  | 0.826 | 0.954 |
| GPT-3.5 | 5%                    | Human Validation | 0.932 | 0.008 | 0.932  | 0.923 | 0.944 |
| GPT-3.5 | 5%                    | Debiased         | 0.938 | 0.008 | 0.938  | 0.927 | 0.951 |
| GPT-3.5 | 10%                   | LLM              | 0.915 | 0.036 | 0.926  | 0.853 | 0.950 |
| GPT-3.5 | 10%                   | Human Validation | 0.940 | 0.008 | 0.942  | 0.926 | 0.953 |
| GPT-3.5 | 10%                   | Debiased         | 0.943 | 0.008 | 0.944  | 0.930 | 0.956 |
| GPT-3.5 | 25%                   | LLM              | 0.913 | 0.039 | 0.923  | 0.832 | 0.950 |
| GPT-3.5 | 25%                   | Human Validation | 0.946 | 0.007 | 0.946  | 0.934 | 0.959 |
| GPT-3.5 | 25%                   | Debiased         | 0.947 | 0.005 | 0.947  | 0.939 | 0.955 |
| GPT-3.5 | 50%                   | LLM              | 0.915 | 0.037 | 0.921  | 0.836 | 0.954 |
| GPT-3.5 | 50%                   | Human Validation | 0.947 | 0.005 | 0.945  | 0.939 | 0.954 |
| GPT-3.5 | 50%                   | Debiased         | 0.948 | 0.006 | 0.947  | 0.936 | 0.958 |
| GPT-4o  | 5%                    | LLM              | 0.936 | 0.019 | 0.940  | 0.892 | 0.957 |
| GPT-4o  | 5%                    | Human Validation | 0.932 | 0.010 | 0.931  | 0.916 | 0.947 |
| GPT-4o  | 5%                    | Debiased         | 0.927 | 0.009 | 0.926  | 0.912 | 0.941 |
| GPT-4o  | 10%                   | LLM              | 0.937 | 0.017 | 0.943  | 0.903 | 0.953 |
| GPT-4o  | 10%                   | Human Validation | 0.940 | 0.007 | 0.940  | 0.928 | 0.948 |
| GPT-4o  | 10%                   | Debiased         | 0.939 | 0.009 | 0.940  | 0.925 | 0.952 |
| GPT-4o  | 25%                   | LLM              | 0.937 | 0.019 | 0.942  | 0.901 | 0.958 |
| GPT-4o  | 25%                   | Human Validation | 0.947 | 0.008 | 0.947  | 0.934 | 0.956 |
| GPT-4o  | 25%                   | Debiased         | 0.946 | 0.008 | 0.946  | 0.933 | 0.958 |
| GPT-4o  | 50%                   | LLM              | 0.937 | 0.019 | 0.943  | 0.895 | 0.956 |
| GPT-4o  | 50%                   | Human Validation | 0.947 | 0.006 | 0.948  | 0.938 | 0.959 |
| GPT-4o  | 50%                   | Debiased         | 0.945 | 0.007 | 0.948  | 0.933 | 0.955 |

Table 54: Coverage Summary Statistics of  $\beta_{19}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|-------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.865 | 0.078 | 0.899  | 0.715 | 0.948 |
| GPT-3.5 | 5%                    | Human Validation | 0.937 | 0.007 | 0.938  | 0.928 | 0.947 |
| GPT-3.5 | 5%                    | Debiased         | 0.945 | 0.008 | 0.946  | 0.932 | 0.956 |
| GPT-3.5 | 10%                   | LLM              | 0.866 | 0.078 | 0.897  | 0.719 | 0.944 |
| GPT-3.5 | 10%                   | Human Validation | 0.943 | 0.009 | 0.943  | 0.930 | 0.957 |
| GPT-3.5 | 10%                   | Debiased         | 0.945 | 0.009 | 0.946  | 0.934 | 0.957 |
| GPT-3.5 | 25%                   | LLM              | 0.868 | 0.078 | 0.893  | 0.715 | 0.949 |
| GPT-3.5 | 25%                   | Human Validation | 0.948 | 0.007 | 0.948  | 0.938 | 0.958 |
| GPT-3.5 | 25%                   | Debiased         | 0.948 | 0.008 | 0.948  | 0.936 | 0.960 |
| GPT-3.5 | 50%                   | LLM              | 0.864 | 0.081 | 0.895  | 0.717 | 0.943 |
| GPT-3.5 | 50%                   | Human Validation | 0.951 | 0.007 | 0.950  | 0.940 | 0.963 |
| GPT-3.5 | 50%                   | Debiased         | 0.946 | 0.008 | 0.946  | 0.934 | 0.958 |
| GPT-4o  | 5%                    | LLM              | 0.917 | 0.034 | 0.927  | 0.853 | 0.957 |
| GPT-4o  | 5%                    | Human Validation | 0.938 | 0.010 | 0.939  | 0.917 | 0.953 |
| GPT-4o  | 5%                    | Debiased         | 0.942 | 0.009 | 0.941  | 0.928 | 0.956 |
| GPT-4o  | 10%                   | LLM              | 0.914 | 0.038 | 0.928  | 0.848 | 0.950 |
| GPT-4o  | 10%                   | Human Validation | 0.943 | 0.007 | 0.942  | 0.932 | 0.952 |
| GPT-4o  | 10%                   | Debiased         | 0.948 | 0.008 | 0.948  | 0.935 | 0.958 |
| GPT-4o  | 25%                   | LLM              | 0.919 | 0.030 | 0.929  | 0.862 | 0.956 |
| GPT-4o  | 25%                   | Human Validation | 0.946 | 0.007 | 0.945  | 0.936 | 0.956 |
| GPT-4o  | 25%                   | Debiased         | 0.947 | 0.008 | 0.947  | 0.935 | 0.958 |
| GPT-4o  | 50%                   | LLM              | 0.914 | 0.031 | 0.921  | 0.860 | 0.950 |
| GPT-4o  | 50%                   | Human Validation | 0.951 | 0.006 | 0.950  | 0.941 | 0.960 |
| GPT-4o  | 50%                   | Debiased         | 0.948 | 0.005 | 0.948  | 0.938 | 0.954 |

Table 55: Coverage Summary Statistics of  $\beta_{20}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|-------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.913 | 0.053 | 0.930  | 0.822 | 0.956 |
| GPT-3.5 | 5%                    | Human Validation | 0.938 | 0.006 | 0.938  | 0.930 | 0.946 |
| GPT-3.5 | 5%                    | Debiased         | 0.941 | 0.007 | 0.942  | 0.929 | 0.953 |
| GPT-3.5 | 10%                   | LLM              | 0.911 | 0.049 | 0.927  | 0.813 | 0.953 |
| GPT-3.5 | 10%                   | Human Validation | 0.942 | 0.008 | 0.941  | 0.932 | 0.956 |
| GPT-3.5 | 10%                   | Debiased         | 0.944 | 0.008 | 0.941  | 0.934 | 0.958 |
| GPT-3.5 | 25%                   | LLM              | 0.912 | 0.051 | 0.930  | 0.821 | 0.952 |
| GPT-3.5 | 25%                   | Human Validation | 0.946 | 0.007 | 0.946  | 0.937 | 0.956 |
| GPT-3.5 | 25%                   | Debiased         | 0.948 | 0.006 | 0.948  | 0.938 | 0.960 |
| GPT-3.5 | 50%                   | LLM              | 0.912 | 0.050 | 0.929  | 0.818 | 0.955 |
| GPT-3.5 | 50%                   | Human Validation | 0.950 | 0.006 | 0.951  | 0.941 | 0.959 |
| GPT-3.5 | 50%                   | Debiased         | 0.948 | 0.008 | 0.946  | 0.937 | 0.963 |
| GPT-4o  | 5%                    | LLM              | 0.889 | 0.068 | 0.902  | 0.796 | 0.952 |
| GPT-4o  | 5%                    | Human Validation | 0.937 | 0.009 | 0.937  | 0.924 | 0.950 |
| GPT-4o  | 5%                    | Debiased         | 0.936 | 0.008 | 0.936  | 0.924 | 0.947 |
| GPT-4o  | 10%                   | LLM              | 0.893 | 0.062 | 0.915  | 0.794 | 0.953 |
| GPT-4o  | 10%                   | Human Validation | 0.944 | 0.006 | 0.943  | 0.935 | 0.955 |
| GPT-4o  | 10%                   | Debiased         | 0.945 | 0.006 | 0.946  | 0.934 | 0.955 |
| GPT-4o  | 25%                   | LLM              | 0.893 | 0.063 | 0.914  | 0.792 | 0.951 |
| GPT-4o  | 25%                   | Human Validation | 0.949 | 0.007 | 0.950  | 0.937 | 0.962 |
| GPT-4o  | 25%                   | Debiased         | 0.946 | 0.007 | 0.945  | 0.936 | 0.956 |
| GPT-4o  | 50%                   | LLM              | 0.895 | 0.060 | 0.917  | 0.796 | 0.951 |
| GPT-4o  | 50%                   | Human Validation | 0.950 | 0.008 | 0.950  | 0.936 | 0.963 |
| GPT-4o  | 50%                   | Debiased         | 0.949 | 0.007 | 0.949  | 0.940 | 0.959 |

Table 56: Coverage Summary Statistics of  $\beta_{\text{Other}}$  by Model and Proportion of Validation Samples. Proxy on the RHS:  $V = Y^\top \beta = Y_3\beta_3 + Y_{14}\beta_{14} + Y_{15}\beta_{15} + Y_{19}\beta_{19} + Y_{20}\beta_{20} + Y_{\text{Other}}\beta_{\text{Other}}$

| Model   | Validation Proportion | MSE              | Mean  | SD    | Median | 5%    | 95%   |
|---------|-----------------------|------------------|-------|-------|--------|-------|-------|
| GPT-3.5 | 5%                    | LLM              | 0.769 | 0.156 | 0.797  | 0.503 | 0.944 |
| GPT-3.5 | 5%                    | Human Validation | 0.950 | 0.007 | 0.950  | 0.938 | 0.963 |
| GPT-3.5 | 5%                    | Debiased         | 0.956 | 0.007 | 0.957  | 0.947 | 0.969 |
| GPT-3.5 | 10%                   | LLM              | 0.773 | 0.157 | 0.805  | 0.508 | 0.942 |
| GPT-3.5 | 10%                   | Human Validation | 0.951 | 0.007 | 0.951  | 0.940 | 0.961 |
| GPT-3.5 | 10%                   | Debiased         | 0.952 | 0.007 | 0.952  | 0.941 | 0.963 |
| GPT-3.5 | 25%                   | LLM              | 0.774 | 0.156 | 0.804  | 0.502 | 0.947 |
| GPT-3.5 | 25%                   | Human Validation | 0.950 | 0.007 | 0.951  | 0.938 | 0.959 |
| GPT-3.5 | 25%                   | Debiased         | 0.948 | 0.007 | 0.948  | 0.935 | 0.960 |
| GPT-3.5 | 50%                   | LLM              | 0.774 | 0.151 | 0.796  | 0.522 | 0.942 |
| GPT-3.5 | 50%                   | Human Validation | 0.950 | 0.006 | 0.951  | 0.940 | 0.960 |
| GPT-3.5 | 50%                   | Debiased         | 0.948 | 0.008 | 0.949  | 0.936 | 0.961 |
| GPT-4o  | 5%                    | LLM              | 0.903 | 0.047 | 0.919  | 0.813 | 0.956 |
| GPT-4o  | 5%                    | Human Validation | 0.947 | 0.007 | 0.947  | 0.938 | 0.959 |
| GPT-4o  | 5%                    | Debiased         | 0.952 | 0.007 | 0.952  | 0.943 | 0.965 |
| GPT-4o  | 10%                   | LLM              | 0.908 | 0.046 | 0.928  | 0.829 | 0.956 |
| GPT-4o  | 10%                   | Human Validation | 0.949 | 0.007 | 0.950  | 0.938 | 0.958 |
| GPT-4o  | 10%                   | Debiased         | 0.950 | 0.008 | 0.952  | 0.936 | 0.962 |
| GPT-4o  | 25%                   | LLM              | 0.905 | 0.049 | 0.923  | 0.819 | 0.951 |
| GPT-4o  | 25%                   | Human Validation | 0.948 | 0.006 | 0.949  | 0.939 | 0.957 |
| GPT-4o  | 25%                   | Debiased         | 0.948 | 0.008 | 0.951  | 0.935 | 0.958 |
| GPT-4o  | 50%                   | LLM              | 0.907 | 0.045 | 0.920  | 0.832 | 0.958 |
| GPT-4o  | 50%                   | Human Validation | 0.951 | 0.007 | 0.949  | 0.941 | 0.964 |
| GPT-4o  | 50%                   | Debiased         | 0.950 | 0.006 | 0.949  | 0.941 | 0.959 |